YAKEEN

NEET 2.0 2027

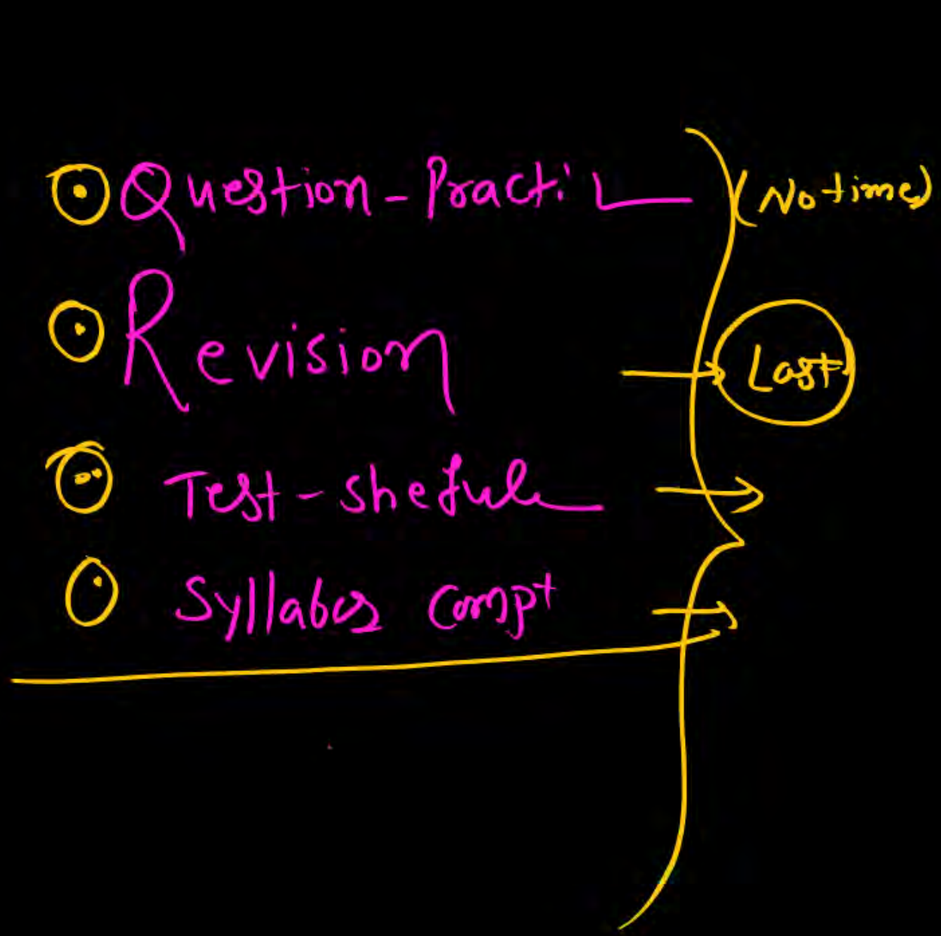
Physics | Lecture - 01

Basic Maths and Calculus

By- Manish Raj (MR Sir)



Topics to be covered



Trigonometry

PHYSICS

Patience
(Basic se concept se sikho, samjo)

Pen - use karo notes banao

Paper
Ls rough page me likh-likh ke practic karo
Achi R..

Practice
↓
do DPP / Assignment

Progress

RE-NEET-2026

Physics - 43/45





Telegm chu

Lecture duration → 2 hr

Lecture → Basic concept feel ke sath to PhD on advance level.

→ do all lecture (dont miss)

→ make Notes in class

→ Sare Question Nahi likhna hai (rough Page pe karna)

→ Jo Question 1st time me Nahi huaa mark karo notes me

→ language wale class me karenge + H/W + Assignment.

→ After class rough Page me sare concept likh ke Practic karo.

→ sare relevent / NEET syllabus + NCERT" Pahle cover karenge.

→ NEET-/JEE mains/ Advance/ PYQ cover in class.

→ Daily Target HOME/work

* 1. Speed - accuracy
2. language Based Question
3. graph Based Question
4. lengthy Calculation
5. theory Based Question
6. NCERT - Question
7. True-false - A/R Based
8. Paragraph Based Subjective Prob^{m}.

for Question Practice [Rough Page]

Sangharsh Assignment → 109 + (40 new) = 150
(graph + calculative + language lengthy)

mahamanthan (NCERT + Theory) (28)

NCERT (Back-exercise) / NCERT - Exampler

Books:-

MR. Physics (all Question of class)

~~Physics~~ med easy → only 11th

Question on new language → HCV question (class + assignment)

⇒ Complete Revision In class

⇒ 50 extra Test Papers + (New lang Based)

YE PAP MAT KARNA

(1) Ye to easy hai, class nahi karna PDF se ho jayega

(2) Bina proper question read kiye answer dena (option read Kariye)

(3) Half calculation ke bad sochna, ho to gya isse ye aa jayega
(Bio me direct answer dikh jata hai

(4) H/W / question practice nahi kar rha, (sirf teacher ke question solve karne se nahi hoga)

(5) Direct tough question karna hai, pahle basic ka master hona hoga.

(6) Backlog hai to test nahi diya

(7) Life me dosti, pyar bhi jaruri hai.

(8) Fresh hone ke liye 20-25 minute to insta/youtube jaruri hai

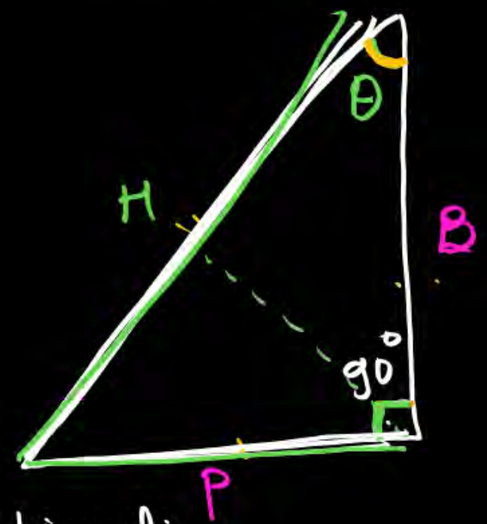
Basic Math

1. Trigonometry
2. Basic calculation
3. graph
4. AIP, G.P.
5. diffrentiation (30) + (H/W 60)
6. maxima / minima
7. Integration
8. log.

Trigonometry

Right-Angle Triangle

Copy



Trigonometric ratio:-

- $\sin\theta = \frac{P}{H}$ = Pure number, Unit & dimensionless
- $\cos\theta = \frac{B}{H}$
- $\tan\theta = \frac{P}{B}$
- $\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{H}{P}$
- $\sec\theta = \frac{1}{\cos\theta} = \frac{H}{B}$
- $\cot\theta = \frac{1}{\tan\theta} = \frac{B}{P}$

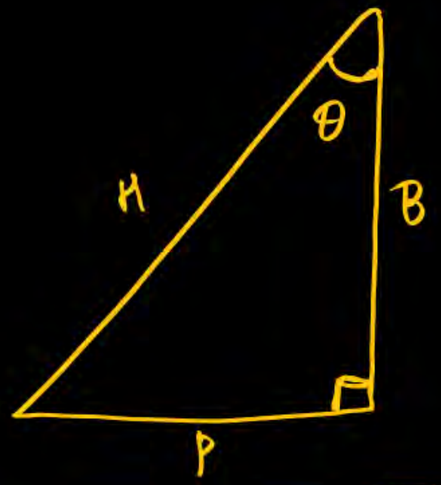
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$$\sec\theta = \frac{1}{\cos\theta}$$

$\sec\theta \cdot \cos\theta = 1$

$\sin\theta \cdot \operatorname{cosec}\theta = 1$

$\tan\theta \cdot \cot\theta = 1$



$$B^2 + P^2 = H^2$$

divided by H^2 Both side.

$$\frac{B^2}{H^2} + \frac{P^2}{H^2} = \frac{H^2}{H^2}$$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\cos\theta = \sqrt{1 - \sin^2\theta}$$

$$P^2 + B^2 = H^2$$

divided by P^2 both side

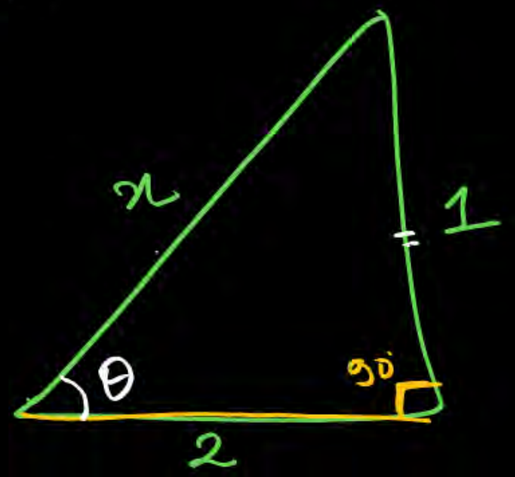
$$\frac{P^2}{P^2} + \frac{B^2}{P^2} = \frac{H^2}{P^2}$$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

divided by B^2

$$\tan^2\theta + 1 = \sec^2\theta$$

① find x and $\sin\theta / \cos\theta$



$$\sin\theta = \frac{1}{\sqrt{5}}$$

$$\cos\theta = \frac{2}{\sqrt{5}}$$

$$\tan\theta = \frac{1}{2}$$

$P^2 + B^2 = H^2$

$2^2 + 1^2 = x^2$

$4 + 1 = x^2$

$\boxed{x = \sqrt{5}}$

(2) If $\tan\theta = 2$ then find Perpendicular, Base & H.
also find $\sin\theta$ & $\cos$

Soln

$$\tan\theta = 2 = \frac{P}{B} \quad \text{(given in ques)}$$

$$\frac{P}{B} = 2 = \frac{2}{1}$$

$B = x$ (let)

$$\frac{P}{x} = 2$$

$$P = 2x$$

$$H^2 = P^2 + B^2$$

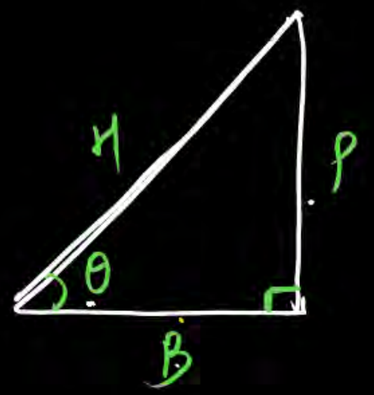
$$H = \sqrt{4x^2 + x^2} = \sqrt{5x^2} = \sqrt{5}\,x$$

$\sin\theta = \frac{P}{H} = \frac{2x}{\sqrt{5}x} = \frac{2}{\sqrt{5}}$

3. If $\sin\theta = 2$ then find value of $\cos\theta$ and $\tan\theta$

Soln → Wrong Question

$$\sin\theta = \frac{P}{H} = 2 \text{ (Not Possible)}$$



$$P^2 + B^2 = H^2$$

$$H > P$$

$$\sin\theta = \frac{P}{H} \leq 1$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$(2)^2 + \cos^2\theta = 1$$

$$\cos^2\theta = 1 - 4$$

$$\cos^2\theta = -3$$

$$\cos\theta = \sqrt{-3}$$ ← Wrong

$$-1 \leq \sin\theta \leq 1$$

$$-1 \leq \cos\theta \leq 1$$

Question copy

(4)

If $\frac{(\sin\theta + \cos\theta)}{(\sin\theta - \cos\theta)} = \frac{11}{5}$ then find value of $\tan\theta$

Soln

$5\sin\theta + 5\cos\theta = 11\sin\theta - 11\cos\theta$

$16\cos\theta = 6\sin\theta$

$\frac{\sin\theta}{\cos\theta} = \frac{16}{6}$

$\boxed{\tan\theta = \frac{8}{3}}$



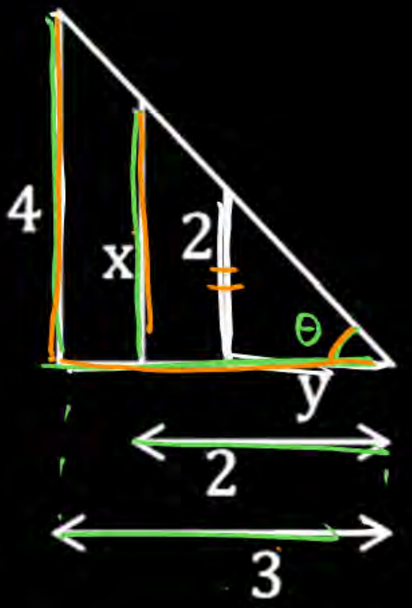
4/3

Question

(5) copy

Find x and y:

1. $x = \frac{4}{3}, y = \frac{3}{2}$
2. $x = \frac{8}{3}, y = \frac{6}{2}$
3. $x = \frac{8}{3}, y = \frac{3}{2}$
4. $x = \frac{3}{2}, y = \frac{8}{3}$



$\tan\theta = \frac{2}{y} = \frac{4}{3}$

$y = \frac{6}{4} = \frac{3}{2}$

$\tan\theta = \frac{4}{3} = \frac{x}{2}$

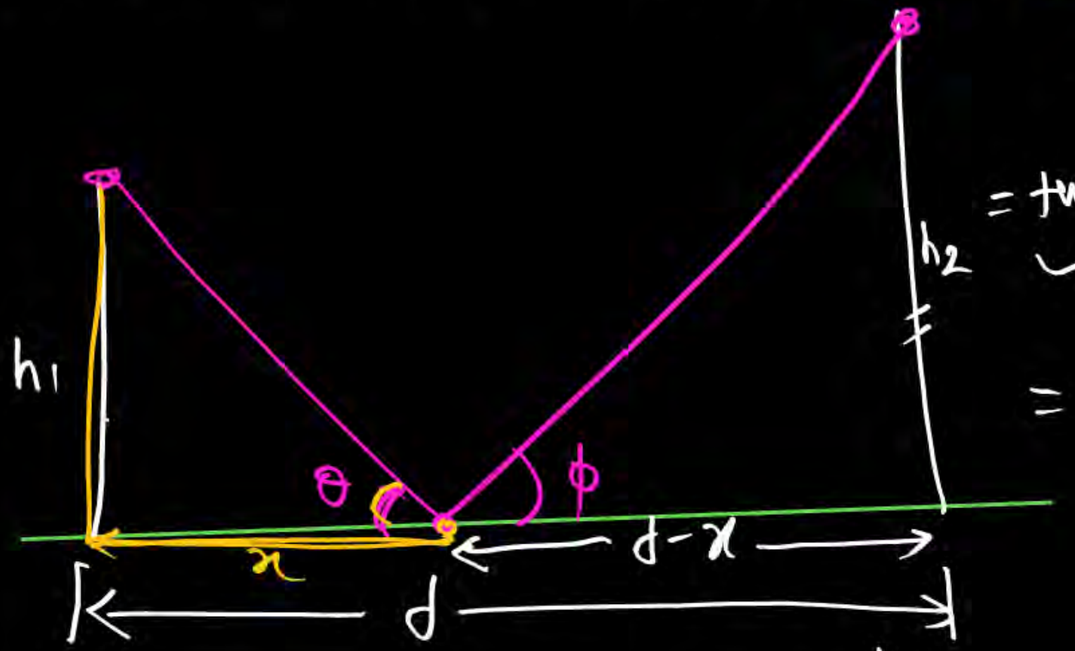
$x = \frac{8}{3}$

Question

copy (ghw)

From a point on level ground between two vertical towers that are d meters apart, the angles of elevation to the tops of the nearer and farther towers are θ and ϕ, respectively. Assuming the observer is x meters from the nearer tower, what is the difference in heights of the towers?

1. $d \,.\, (\tan\theta - \tan\phi)$
2. $x \,.\, \tan\theta - (d - x) \,.\, \tan\phi$
3. $\tan\theta - \tan\phi \,.\, d$
4. $\dfrac{\tan\theta - \tan\phi}{d}$



$h_2 - h_1 = ?$

$= \tan\phi(d-x) - x\tan\theta$

$=$

$\tan\theta = \dfrac{h_1}{x}$

$\tan\phi = \dfrac{h_2}{d-x}$

Question

M/W
No copy (rough)

Two towers are separated by a horizontal distance D. From the top of the first tower (height h_1), the angle of depression to the top of the second tower (height h_2) is θ.

Which relation correctly gives the horizontal separation D in terms of h_1 . h_2, θ?

1. $D = \dfrac{h_1 + h_2}{\tan\theta}$

2. $D = \dfrac{h_1 - h_2}{\tan\theta}$

3. $D = \dfrac{h_2 - h_1}{\tan\theta}$

4. $D = \tan\theta \,.(h_1 + h_2)$

$-1 \le \cos\theta \le 1$

$-1 \le \sin\theta \le 1$

$-\infty \le \tan\theta \le +\infty$

	0°	30°	45°	60°	90°	120°	135	150°	180°
$\sin\theta = \frac{P}{H}$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	$+1$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos\theta$	$+1$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan\theta = \frac{\sin\theta}{\cos\theta}$	0	$\frac{1}{\sqrt{3}}$	1	$(\sqrt{3})$	Undefin (infinity)	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

If θ Increse from 0° to 90°

- $\sin\theta \rightarrow$ Incres
- $\cos\theta \rightarrow$ decres
- $\tan\theta \rightarrow$ Incr

copy

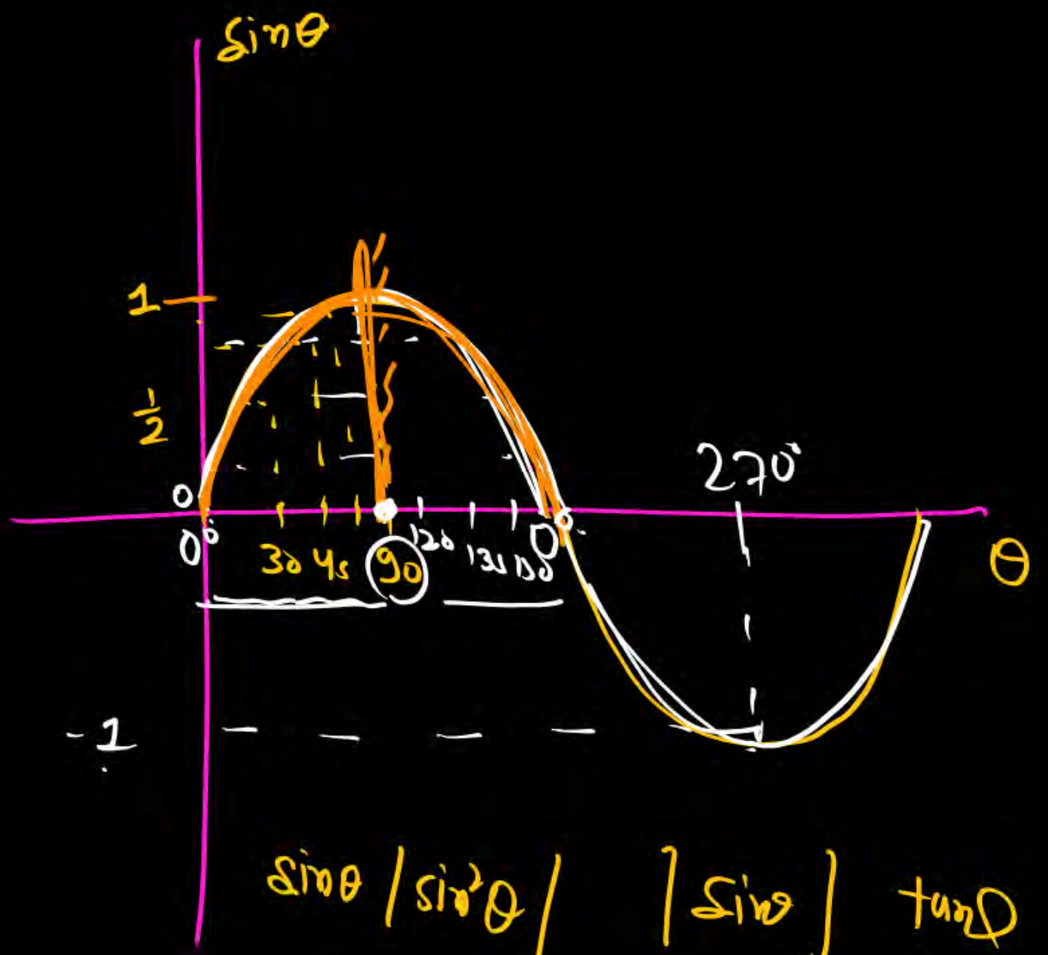
$\sin(120) = \sin(90+30) = \frac{\sqrt{3}}{2}$

$\cos(150^\circ) = -\frac{\sqrt{3}}{2}$

$\tan(135^\circ) = -1$

$$\begin{cases} \sin(150) = \frac{1}{2} \\ \cos(120^\circ) = -\frac{1}{2} \end{cases}$$

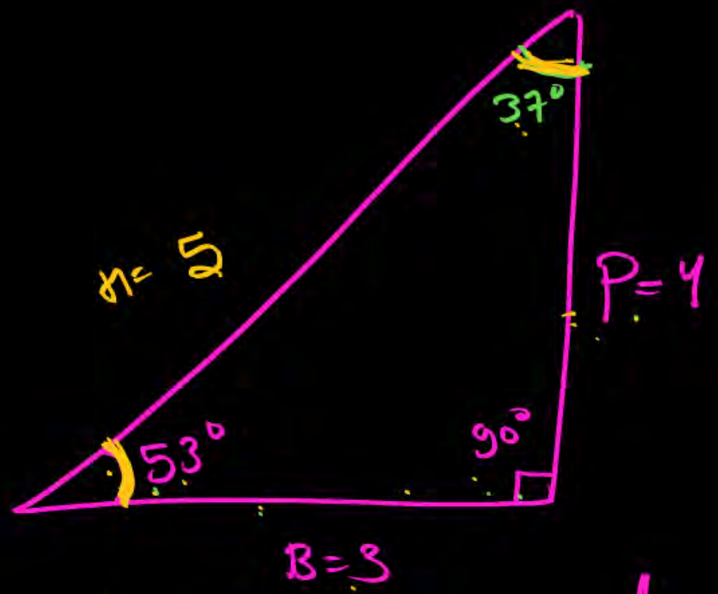
No copy!



sinθ | sin²θ | |sinθ| tanθ

cosθ (ii)θ → Sab ka graph Padhna

Special Angle



$\sin 37^\circ = \frac{3}{5}$

$\cos 37^\circ = \frac{4}{5}$

$\tan(37^\circ) = \frac{3}{4}$

$\tan(53^\circ) = \frac{4}{3}$

$\sin 53^\circ = \frac{4}{5}$

$\cos 53^\circ = \frac{3}{5}$

* $\tan 37^\circ < \tan 53^\circ$

$\left(\frac{3}{4}\right) < \left(\frac{4}{3}\right)$

Question

no copy

If angle increases from 0° to 90°, then value sin θ and cos θ

1. increases, decreases
2. decreases, Increases
3. both decrease
4. both Increase

Question

Copy

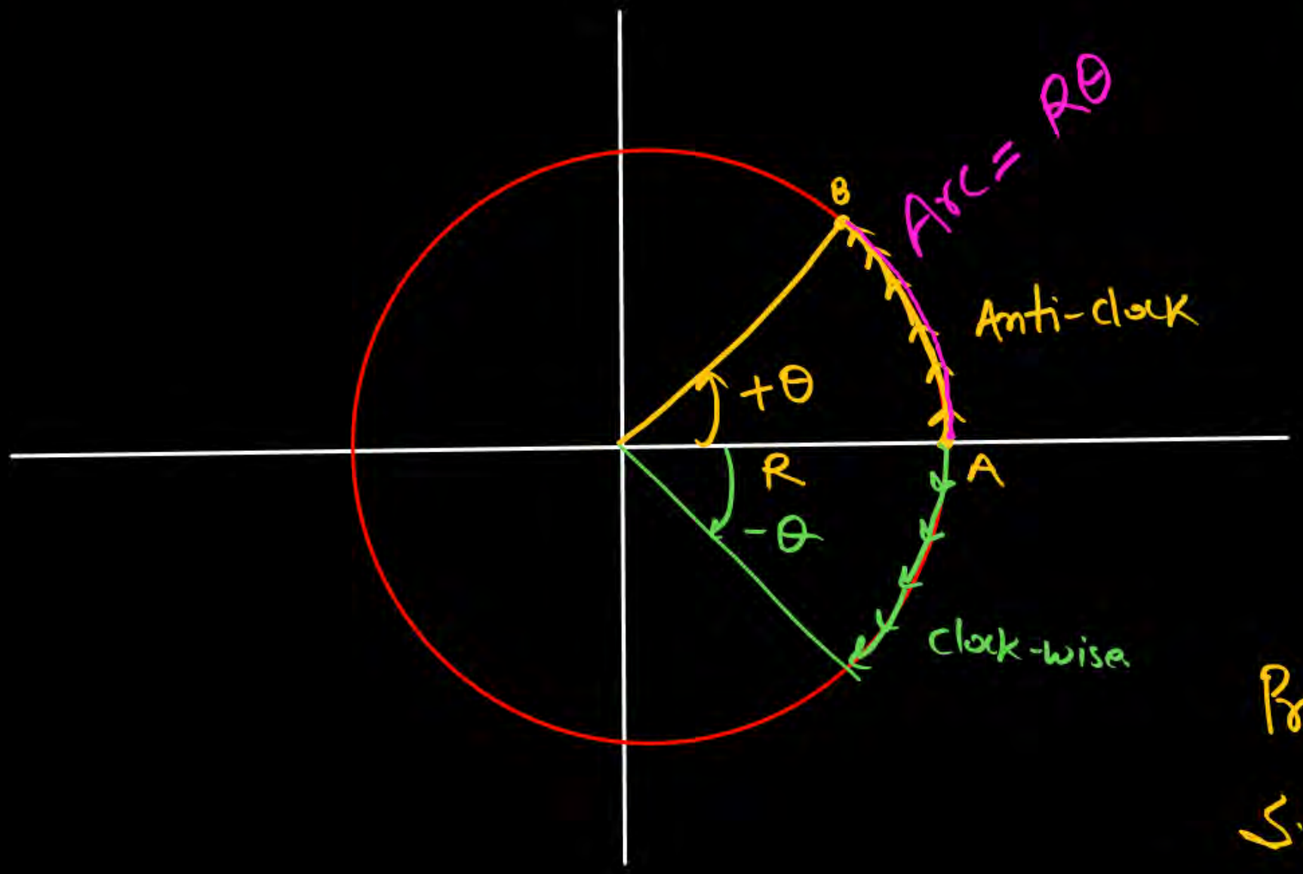
Find value of $\sin^2(40°) + \cos^2(40°) = ??$

$\sin^2(40°) + \cos^2(40°) = ?? = 1$

θ θ

Ansy

Angle and Arc



Angle

$\text{Arc} = R\theta$ ← unit must be in S.I unit radian

$180^\circ = \pi$ rad

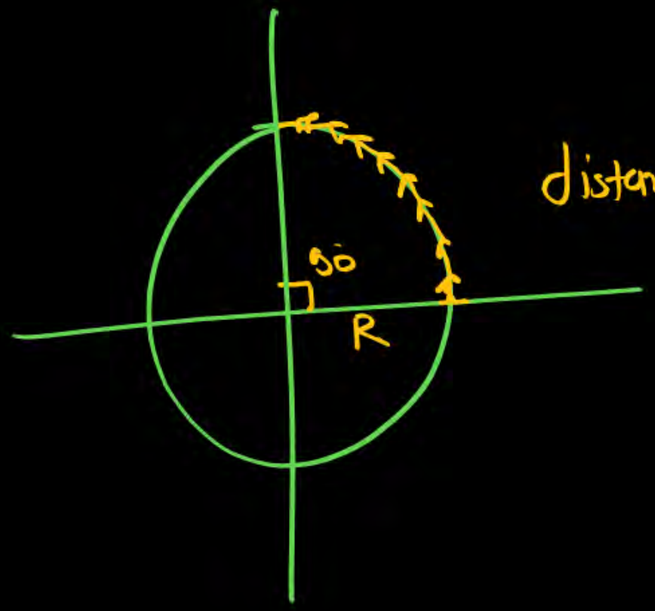
$1^\circ = \frac{\pi}{180}$ rad

$1 \text{ rad} = \frac{180^\circ}{\pi}$

Practical unit of Angle → degree, minitute

S.I unit of Angle is radian.

Q. Object moves on circumference of circle(R) then find total distance moved by object if turns by 90°.



distance = total Path length = Arc = $R\theta$

$= R\,90°$

$Arc = R\left[\frac{\pi}{2}\text{ rad}\right]$

(1) $\frac{\pi}{3}$ rad = ?? (degree)

$\frac{180^{60}}{\cancel{3}_{1}} = 60°$ ✓

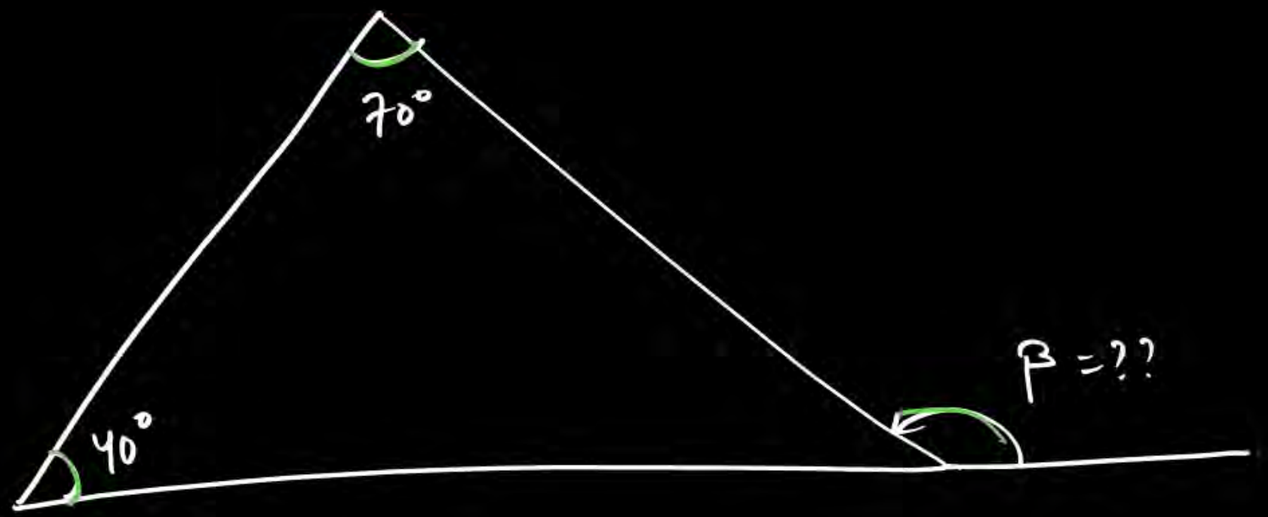
copy

mR ☆ Box

π rad $\rightarrow$ 180° Put

H|w ① find 'β'

copy



Question

some example
Copy in note

H/W

Convert following radian into degree?

(i) $\frac{\pi}{2}$ rad

(ii) $\frac{\pi}{4}$ rad $= \frac{180}{4} = 45$

(iii) $\frac{\pi}{3}$ rad

(iv) $\frac{5\pi}{6}$ rad

(v) $\frac{2\pi}{3}$ rad

(vi) $\frac{4\pi}{3}$ rad

(vii) $\frac{3\pi}{5}$ rad

(viii) 5π rad

(ix) $\frac{2\pi}{5}$ rad $= \frac{2}{5}(180)$

(x) $\frac{7\pi}{12}$ rad

(xi) $\frac{5}{2}\pi$ rad

(xii) $3\ (\pi \text{ rad})$

(xiii) $\frac{\pi}{6}$ rad $= \frac{180}{6} = 30^\circ$

Question

copy some example (HW)

Convert following into radian

(i) $45°$ $= 45\left(\frac{\pi \text{ rad}}{180}\right)$ ✓

(ii) $60°$

(iii) $120°$

(iv) $30°$

(v) $150°$

(vi) $90°$

(vii) π

(viii) $270°$

(ix) $135°$

$180° = \pi$ rad

$1° = \frac{\pi \text{ rad}}{180}$

Question HW No copy

Find value of $\sin(53°) \times \operatorname{cosec}(53°)$

Question

M/w No Cop`

Find value of sin (180°) cos (135°) tan (120°)

Question

H/w No copy

A vertical pole of height h casts a shadow of length l when the Sun is at an angle θ above the horizon.

Which of the following expresses the shadow length l in terms of h and θ?

1. $l = h \,.\, \tan\theta$

2. $l = \dfrac{h}{\tan\theta}$

3. $l = h \,.\, \sin\theta$

4. $l = \dfrac{h}{\sin\theta}$

Question

HW NO Copy

A pole of height h has a wire attached from the top to the ground at an angle θ. If the wire breaks and falls to the ground without changing length, find the horizontal distance x from the base of the pole where the wire touches the ground.

1. $x = h \,.\, \cot\theta$
2. $x = h \,.\, \tan\theta$
3. $x = h \,.\, \sin\theta$
4. $x = h \,.\, \sec\theta$

Question

H/w No copy

A vertical pole of height $h = 10$ m stands on ground that slopes upwards at a constant angle $\alpha = 30°$ with the horizontal. If the sun's angle of elevation above the horizontal is $\theta = 60°$, what is the length of the shadow cast by the pole on the sloping ground?

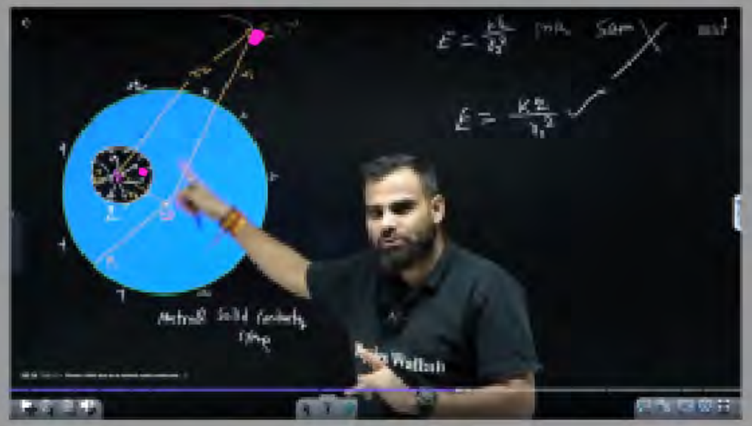
1. 5 m
2. 10 m
3. $10\sqrt{3}$ m
4. $\dfrac{10}{\sqrt{3}-1}$ m

HomE worK rough me solve KarO

DPP Kal aayga

all the best

RE-NEET - 2026



80

21. A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B and E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is :

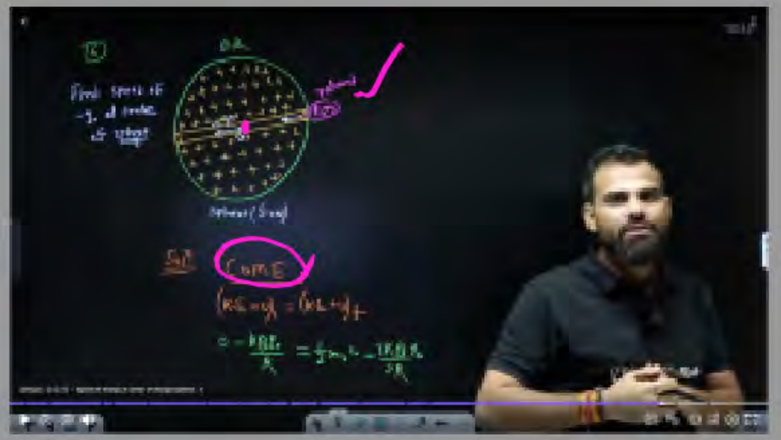


(1) $E_A \neq 0,\ E_B = E_C$

(2) $E_A = 0,\ E_B > E_C$

(3) $E_A \neq 0,\ E_B < E_C$

(4) $E_A = 0,\ E_B = E_C$



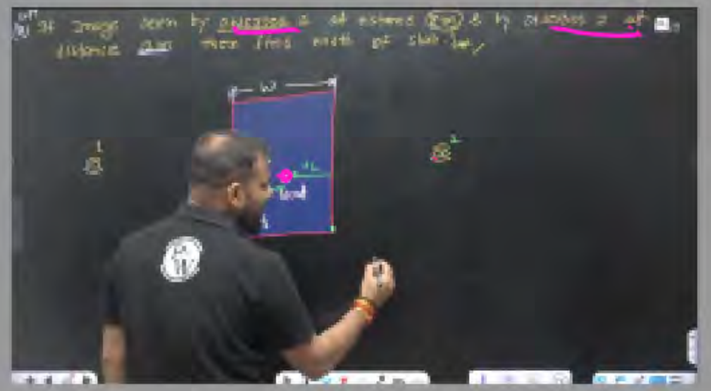
(12)

13. Consider a fixed uniformly charged insulating sphere with radius R and total charge +Q. A point charge −q (q<<Q) with mass m is released from rest at a distance of 3R from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is :

(ϵ_0 is the permittivity of vacuum, neglect gravitational forces).

(1) $\sqrt{\frac{2Qq}{3\pi\epsilon_0 mR}}$ (2) $\sqrt{\frac{Qq}{3\pi\epsilon_0 mR}}$

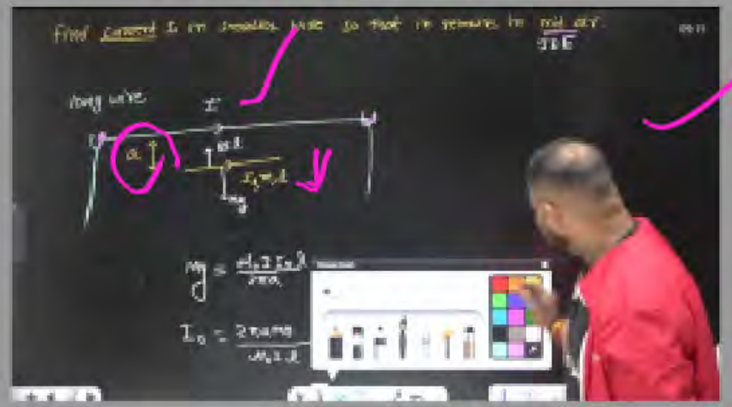
(3) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mR}}$ (4) $\sqrt{\frac{3Qq}{4\pi\epsilon_0 mR}}$



8. Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5, respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the center of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm, is :



(1) 1
(2) 2
(3) 3
(4) 0

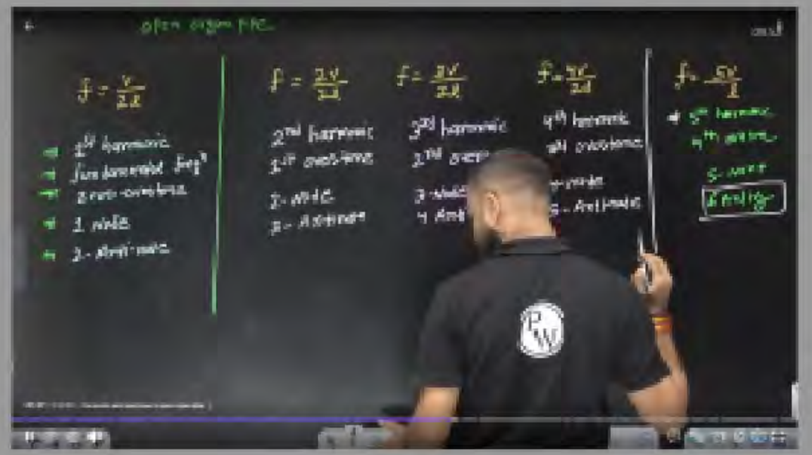


11. Two infinitely long parallel conducting wires A and B carry currents I and 2I, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is :

[g is the acceleration due to gravity and μ_0 is the permeability of free space.]

(1) $\frac{\mu_0 I^2}{\pi\lambda g}$ (2) $\frac{2\mu_0 I^2}{\pi\lambda g}$

(3) $\frac{4\mu_0 I^2}{\pi\lambda g}$ (4) $\frac{\mu_0 I^2}{2\pi\lambda g}$



10. For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is :

(1) $\frac{9}{5}$ (2) 1

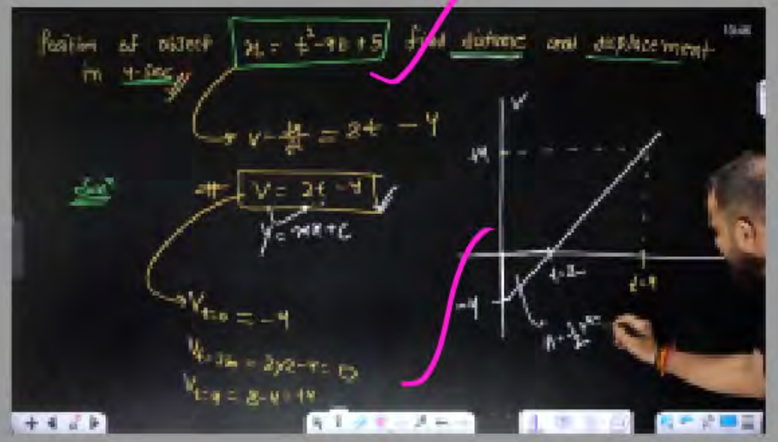
(3) $\frac{3}{5}$ (4) $\frac{5}{9}$



19. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed 10 ms^{-1}, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be 10 ms^{-2}, the value of θ is :

(1) $\tan^{-1}\left(\frac{2}{5}\right)$ (2) $\tan^{-1}\left(\sqrt{3}/2\right)$

(3) $\tan^{-1}\left(2\sqrt{3}\right)$ (4) $\tan^{-1}\left(\frac{1}{5}\right)$



20. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1$ ms^{-2}, $\beta = 6$ ms^{-1} and $\gamma = 5$ m. The average speed of the particle, in ms^{-1}, from $t = 0$ to $t = 6$ s is :

(1) 6 (2) 3

(3) 0 (4) 12

Question 17

Earth has mass 8 times and radius 2 times that of a planet. If the escape velocity from the earth is 11.2 km/s, the escape velocity in km/s from the planet will be:

[28 Jan. 2025 (Shift-II)]

1. 11.2
2. 8.4
3. 5.6
4.

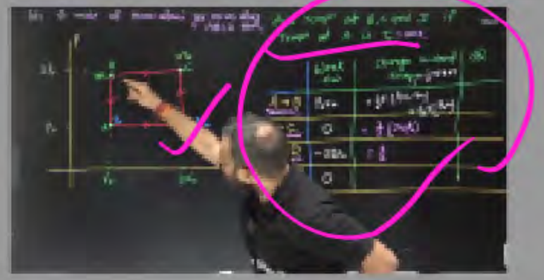
8

3. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2 = \frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is :

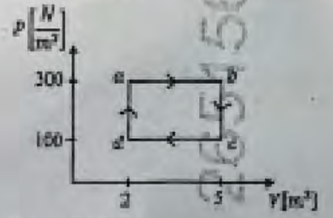
(1) 1 (2) $\sqrt{2}$

(3) 2 (4) $\frac{1}{\sqrt{2}}$





17. One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is :

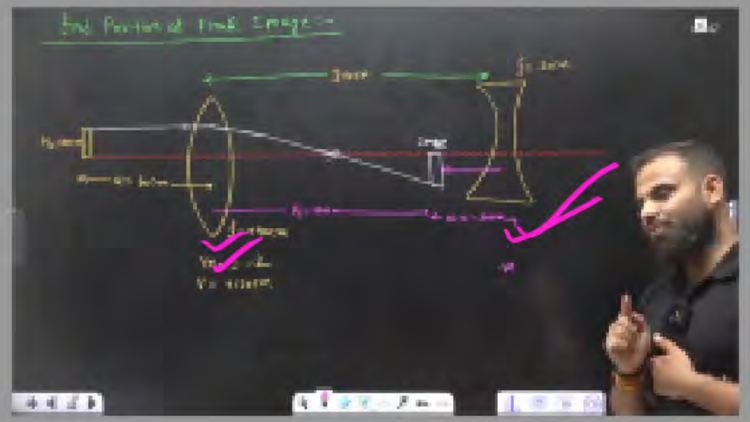


(1) 500 J

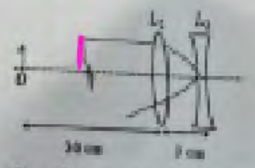
(2) 600 J

(3) 800 J

(4) 400 J



28. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and −10 cm, respectively. The position of the image formed is :

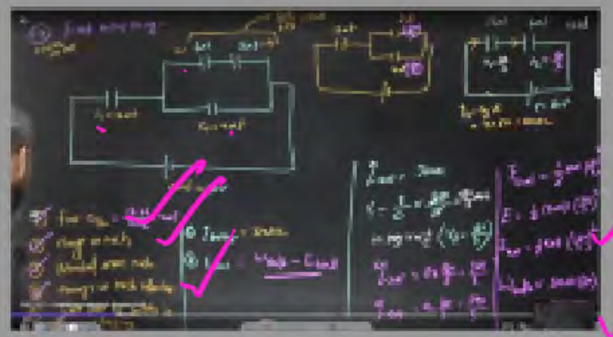


(1) 60 cm to the left of the concave lens

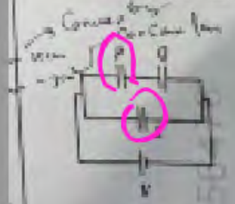
(2) 30 cm to the right of the concave lens

(3) 60 cm to the right of the concave lens

(4) 20 cm to the left of the concave lens



30. Three identical capacitors P, Q and S, each of the capacitance C_0, are connected to a battery of voltage V, as shown in the figure. If the energy stored on the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is :

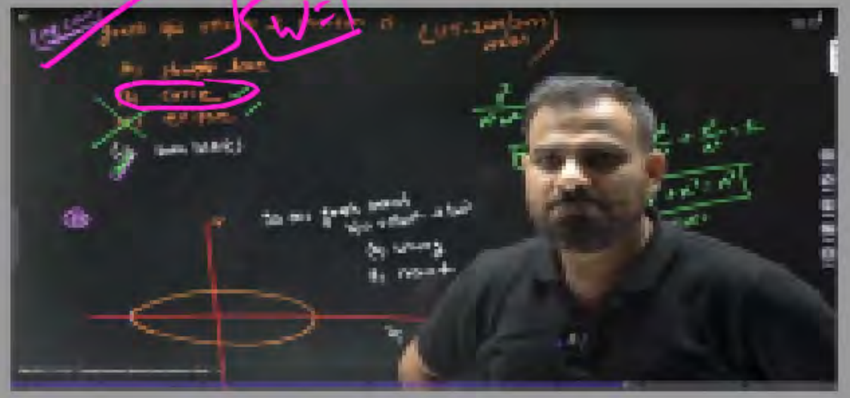


(1) 1/3

(2) 1/2

(3) 1/6

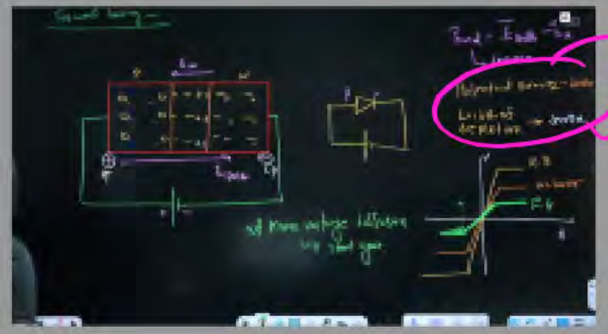
(4) 2/3



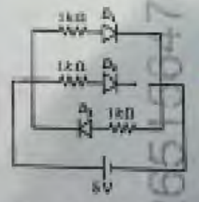
4. Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is k Nm^{-1}. At a given instant the extension of the spring is x metre and the speed of the particle is v ms^{-1}. On the x - v plane, if the graph of v as a function of x is a circle, then the correct option is :

(1) $k = m$ (2) $k = m^2$

(3) $k = \sqrt{m}$ (4) $k = \frac{1}{m}$



3. Three identical p-n junction diodes D_1, D_2 and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is :



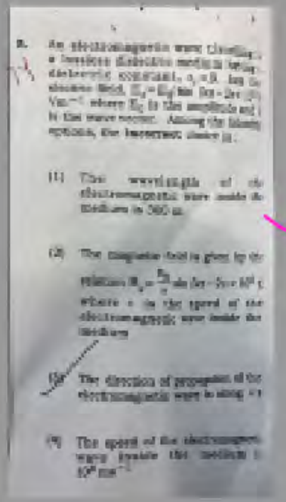
[1] $W_3 = W_1 > W_2$

(2) $W_3 > W_2 > W_1$

(3) $W_2 > W_1 = W_3$

(4) $W_1 > W_2 > W_3$





8. An electromagnetic wave travelling in a lossless dielectric medium having dielectric constant, $\varepsilon_r = 9$. Let its electric field, $E_y = E_0 \sin(kx - 2\pi \times 10^8 t)$ Vm^{-1}, where E_0 is the amplitude and k the wave vector. Among the following options, the incorrect choice is :

(1) The wavelength of the electromagnetic wave inside the medium is 300 m.

(2) The magnetic field is given by the relation $B_z = \frac{E_0}{c} \sin(kx - 2\pi \times 10^8 t)$, where c is the speed of the electromagnetic wave inside the medium.

(3) The direction of propagation of the electromagnetic wave is along +x

(4) The speed of the electromagnetic wave inside the medium is $10^8\ ms^{-1}$



80

18. The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the electromagnetic spectrum		Applications
P	Microwave	I. For purifying the water
Q	UV rays	II. For warming the food
R	Gamma rays	III. For AM and FM communication systems
S	Radio waves	IV. For treating the Cancer cells

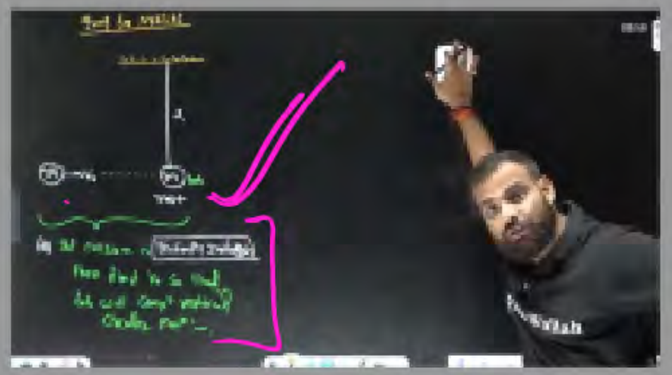
The **correct** option is :

(1) P-I, Q-IV, R-II, S-III

(2) P-II, Q-I, R-IV, S-III

(3) P-II, Q-IV, R-III, S-I

(4) P-I, Q-II, R-III, S-IV

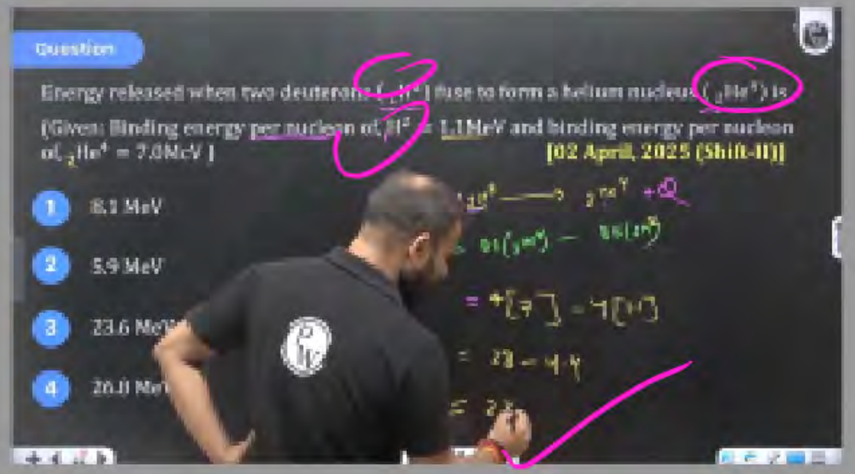


33. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed 10 ms^{-1} hits bob B elastically. The bob B rises h meter after the collision. Taking the acceleration due to gravity $g = 10\ ms^{-2}$ and neglecting the size of the bob, the value of h is :



(1) 7 (2) 5

(3) 2.5 (4) 8



36. Consider the following nuclear reaction :

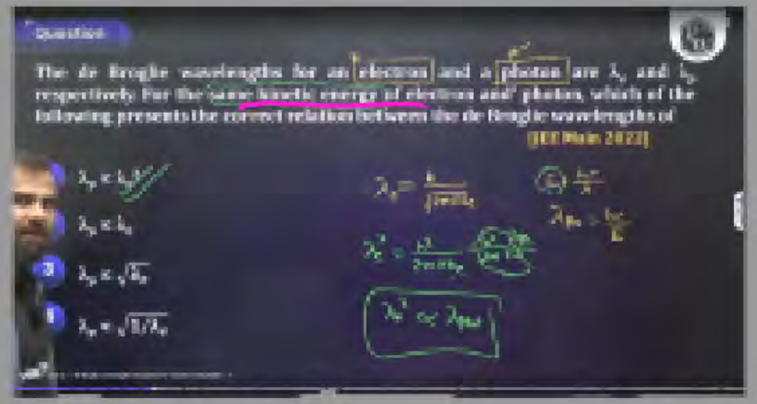
$$^{238}U \rightarrow {}^{234}Th + {}^{4}He$$

Take masses of ^{238}U, ^{234}Th and ^{4}He as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is :

[Given : 1 u = 931.5 MeV c^{-2}]

(1) 3730 (2) 3736

(3) 3740 (4) 3726

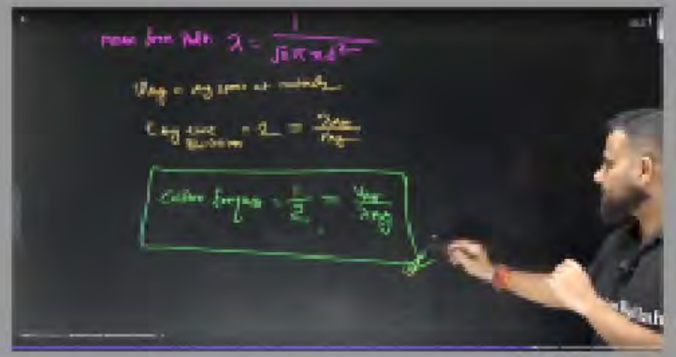


40. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{ph}, $\frac{p_e}{p_{ph}}$ is :

(Take speed of light $= 3 \times 10^8$ ms^{-1}, charge of electron $= -1.6 \times 10^{-19}$ C and mass of electron $= 9 \times 10^{-31}$ kg)

(1) $\frac{1}{250}$ (2) 225

(3) 275 (4) $\frac{2}{450}$



35. The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, then the correct option is :

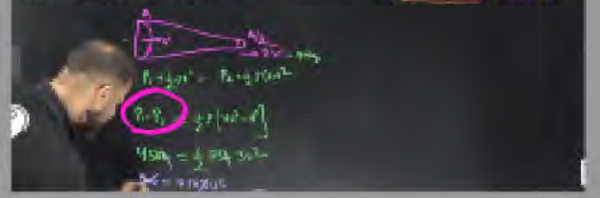
(1) $n_A = 2n_B$ (2) $n_A = \frac{1}{4}n_B$

(3) $n_A = \frac{1}{2}n_B$ (4) $n_A = n_B$

Question

A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe that tapers in cross-sectional area from $A_1 = 1.2 \times 10^{-2}$ m^2 to $A_2 = A_1/2$. The pressure difference between the wide and narrow sections of the pipe is 4500 Pa. The rate of flow of liquid is _____ $\times 10^{-2}$ m^3 s^{-1}.

[28 June, 2022 (Shift-I)]



44. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is 15 Nm^{-2}. The area of cross-section at P and Q are 40 cm^2 and 20 cm^2, respectively. The rate of flow of water through the pipe, in cm^{3}s^{-1}, is :

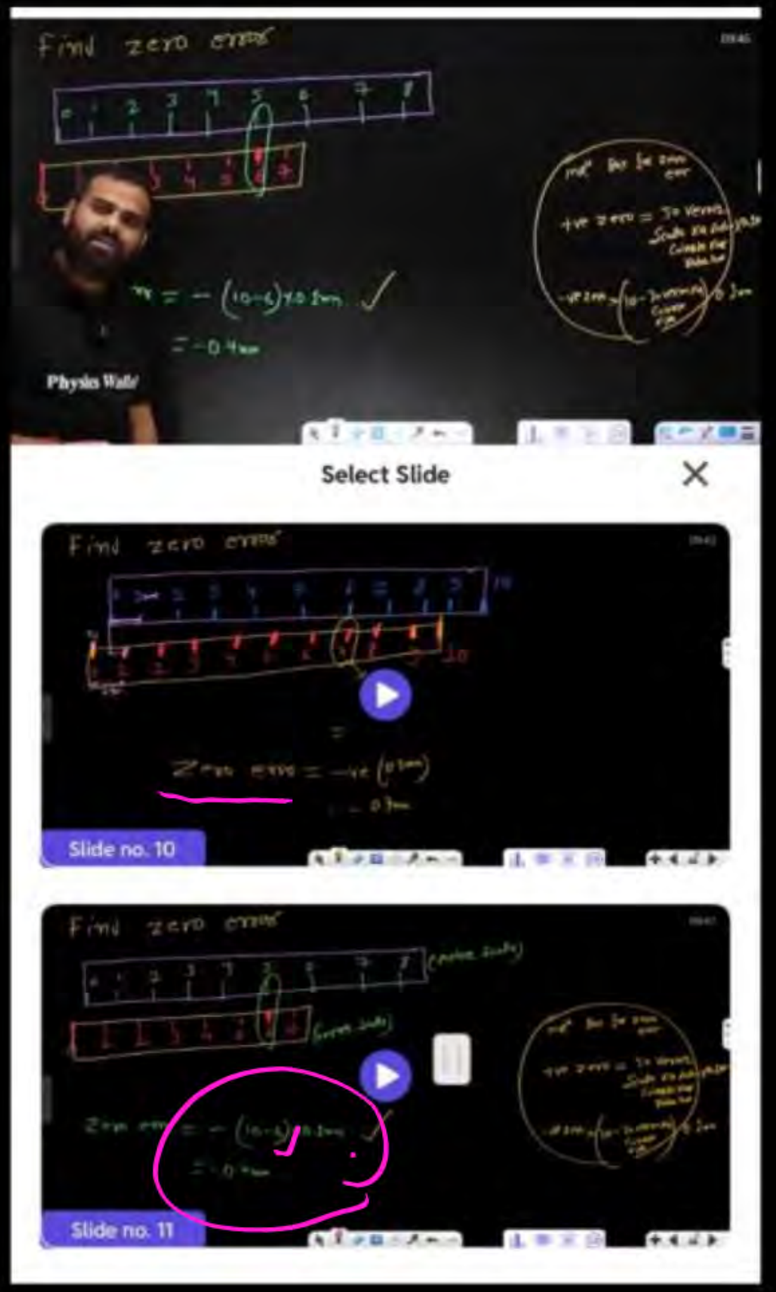
[Take density of water = 1000 kg m^{-3}]



(1) 200 (2) 300

(3) 400 (4) 100

47.



26. One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier caliper measures the length of a wire to be 1 cm, the actual length of the wire is:

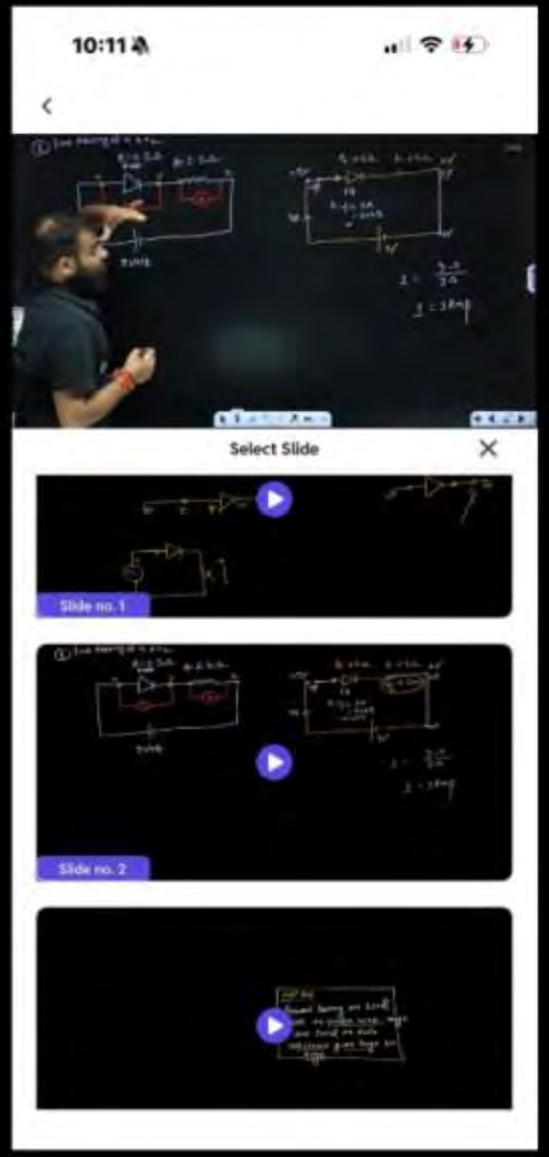
(1) 0.60 cm

(2) 0.96 cm

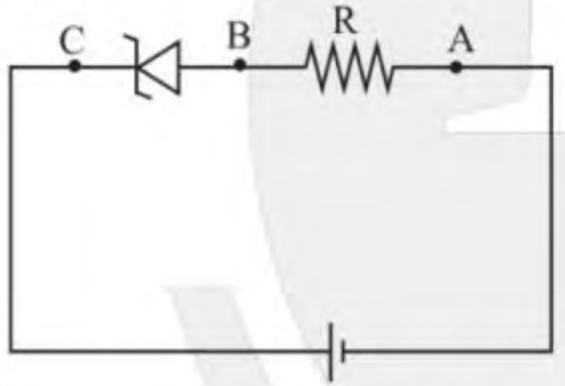
(3) 1.00 cm

(4) 1.04 cm





16. An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage V_i = -5 V. The magnitude of voltage difference between points B and A is :



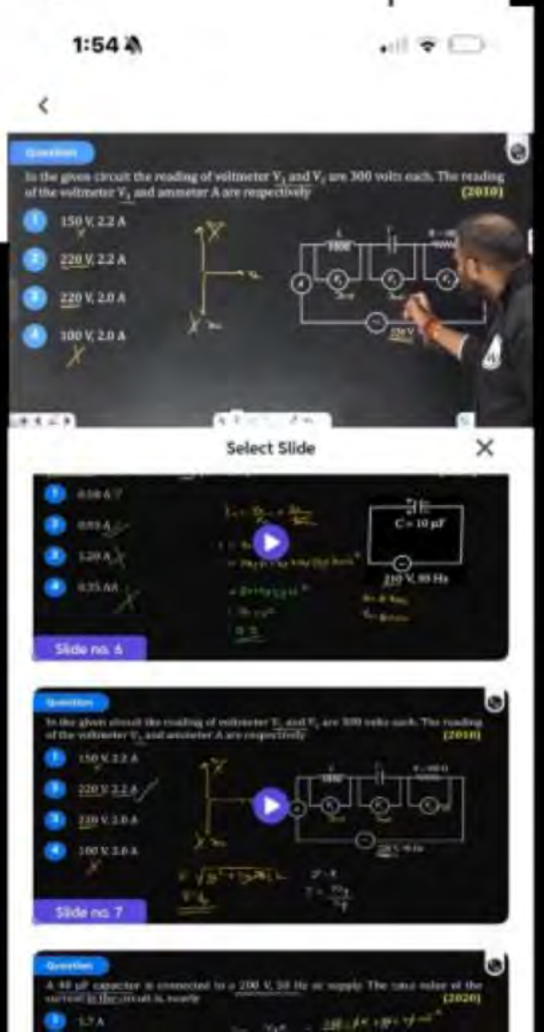
(1) 3 V
(2) 2 V
(3) 1 V
(4) 0 V

30. An ac voltage $V=220\sin(2\times10^3 t)$ Volt is applied to a series LCR circuit. Then the current amplitude in this circuit is :

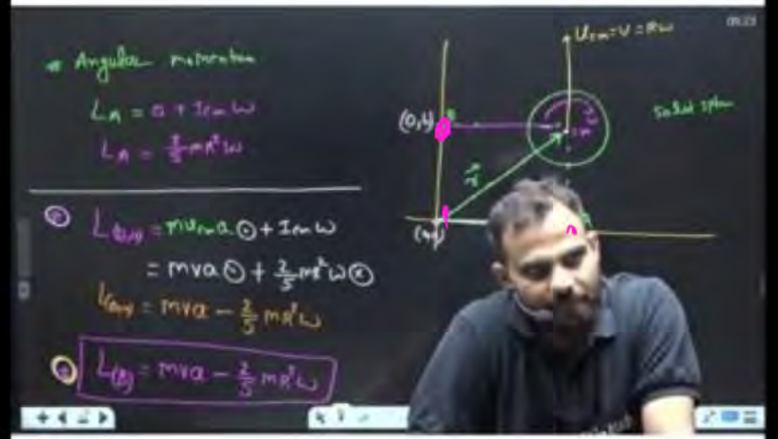
(Given : L=10 mH, C=25μF, R=100 Ω)

(1) 2.2 A (2) 5.5 A

(3) 11.0 A (4) 22.0 A

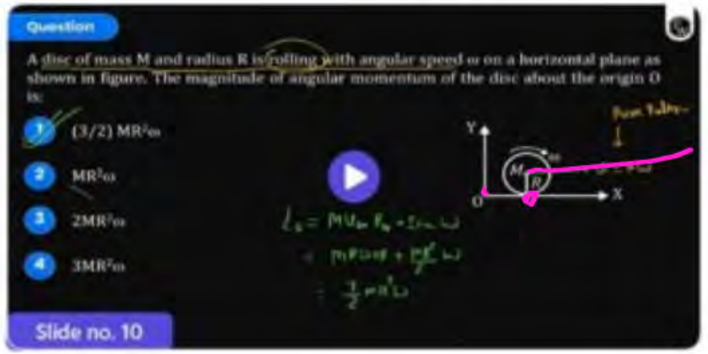


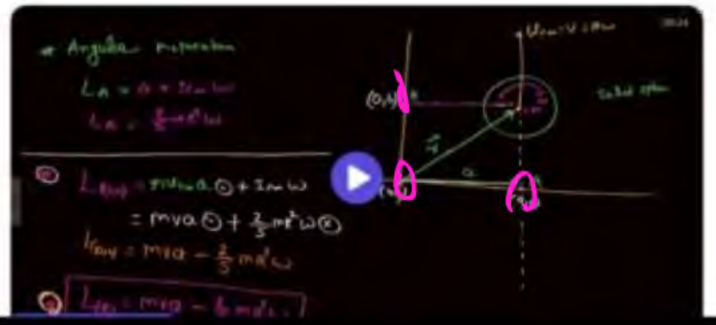




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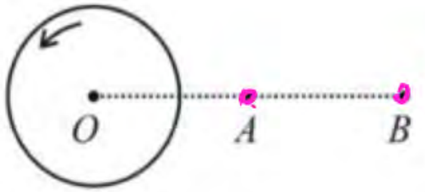






ENGLISH

31. A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2 \times OA$. The value of $\frac{L_A}{L_B}$ is :



(1) $\frac{1}{4}$

(2) $\frac{1}{2}$

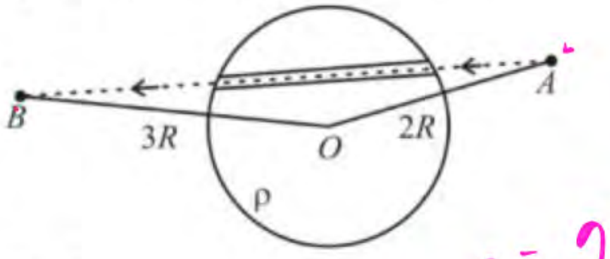
(3) 1

(4) 2

32. A conducting loop of finite resistance lies on the

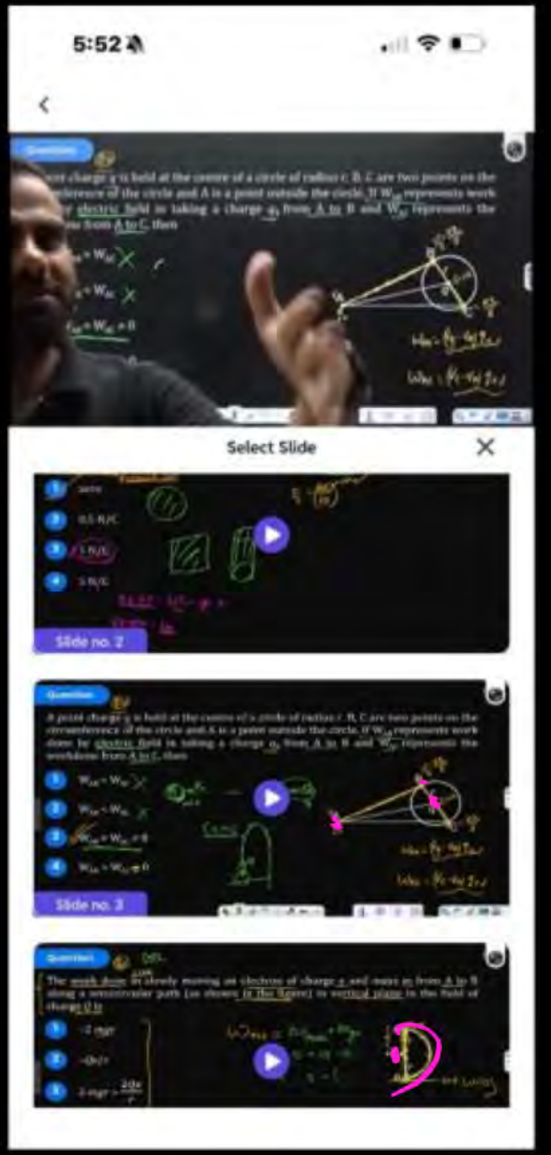
21. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distances $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\epsilon_0}$. The value of n is :

(ϵ_0 is the permittivity of vacuum)

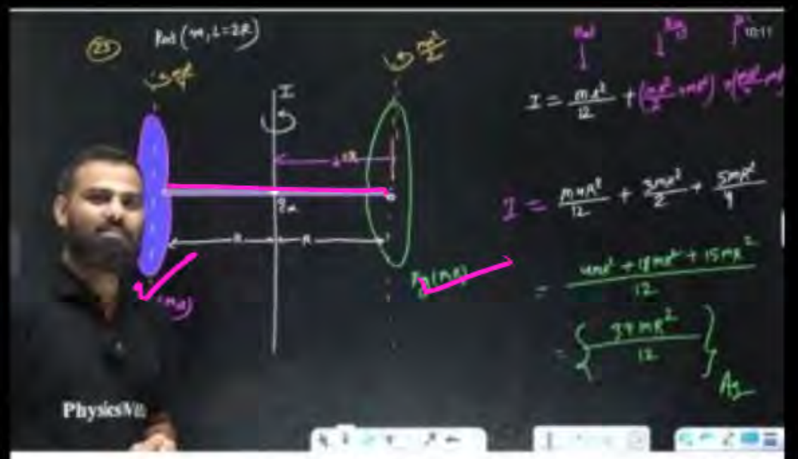


(1) 2
(2) 6
(3) 9
(4) 18

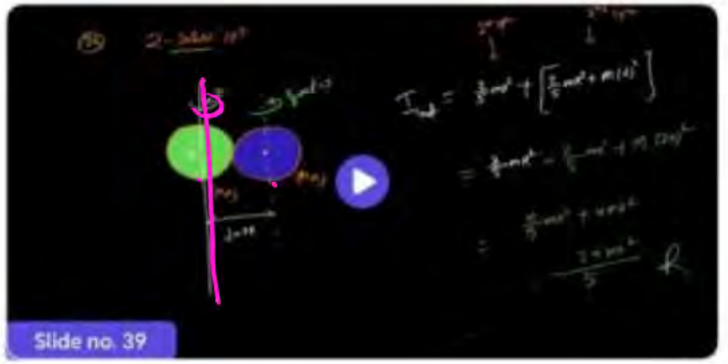
W=20V







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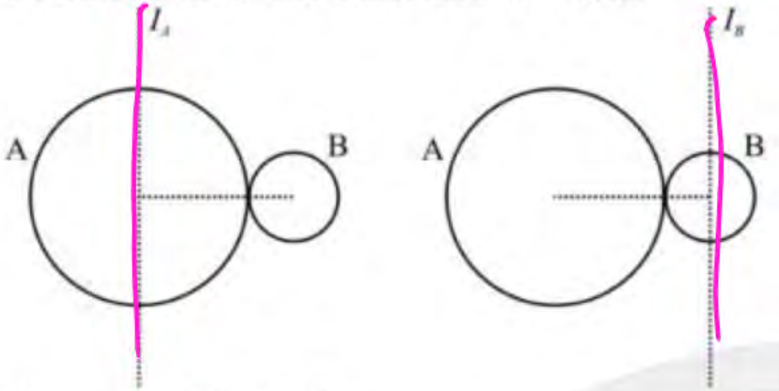


Slide no. 39



Slide no. 40

27. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is:



(1) $(M - m)(R + r)^2$
(2) $(m - M)(R + r)^2$
(3) $(m - M)(R - r)^2$
(4) 0

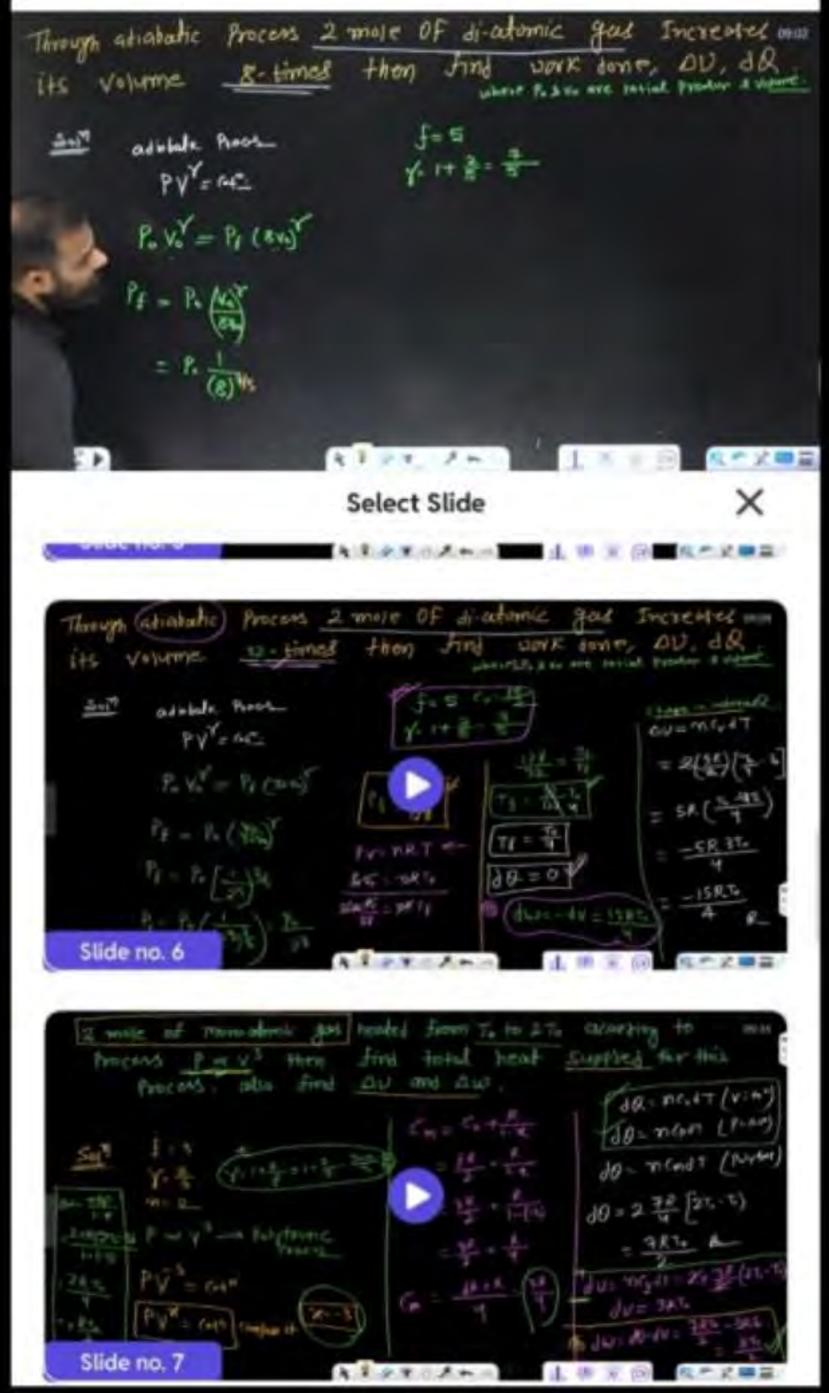
31.

5. Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

(1) $\mu_0 \pi n^2 r^2 l$ (2) $\mu_0 n^2 r^2 l$

(3) $(\mu_0 / 2\pi) n^2 r^2 l$ (4) $2\mu_0 \pi n^2 r^2 l$





17. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60K to 50K. The work done by the gas in the process is :

(Take the universal gas constant as $R = 8.3$ J mol^{-1} K^{-1})

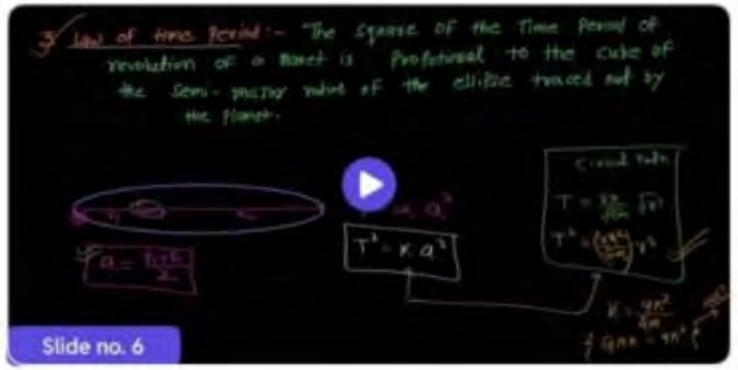
(1) 41.5 J (2) 83 J

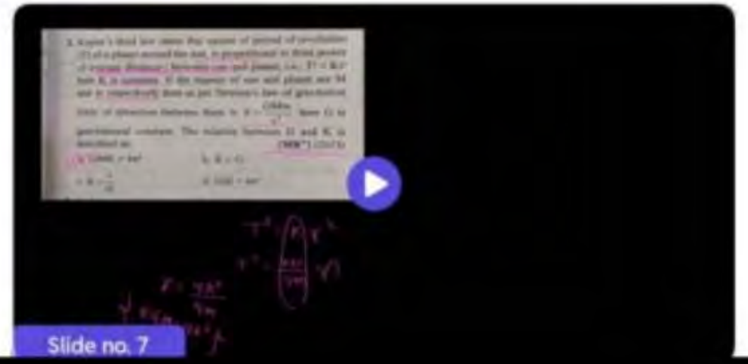
(3) 124.5 J (4) 166 J



18. A ray of light with wavelength λ is incident on three different photo-electric cells namely 1, 2 and 3. The threshold wavelength of these photo-electric cells are λ_1, λ_2 and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ and $\lambda_3 >> \lambda$. The correct option is:



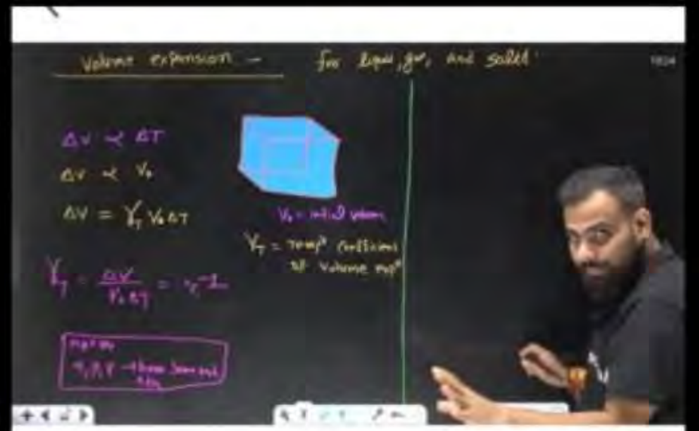




14. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to:

(1) $R^{1/2}$ (2) $R^{3/2}$

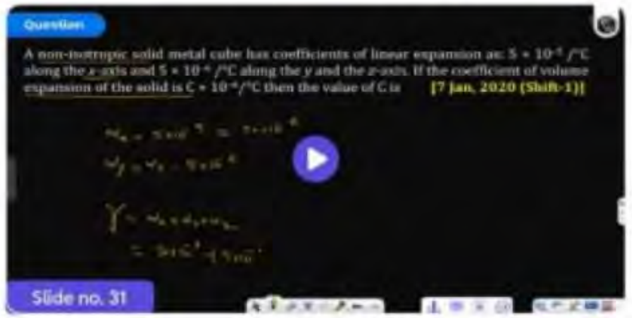
(3) R^2 (4) R^3



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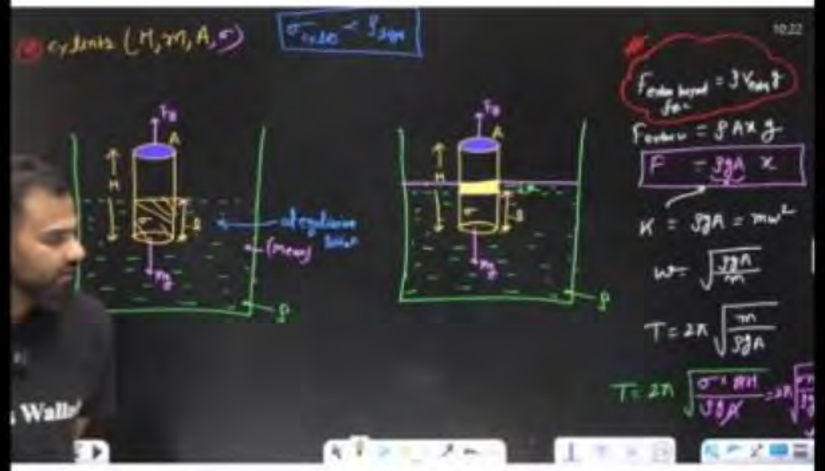
24. The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is :

(1) $2\pi R^3 \alpha \Delta T$

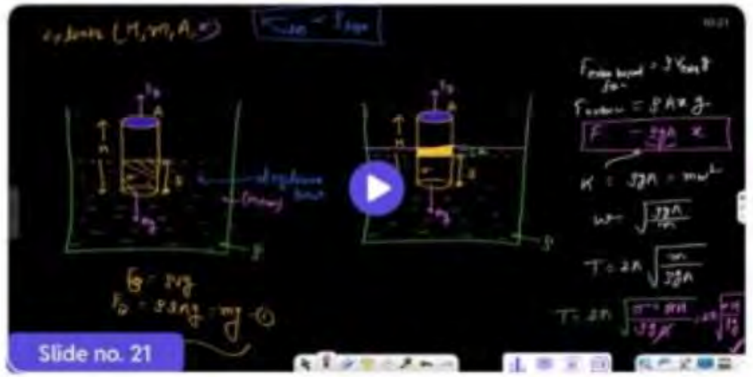
(2) $3\pi R^3 \alpha \Delta T$

(3) $4\pi R^3 \alpha \Delta T$

(4) $6\pi R^3 \alpha \Delta T$



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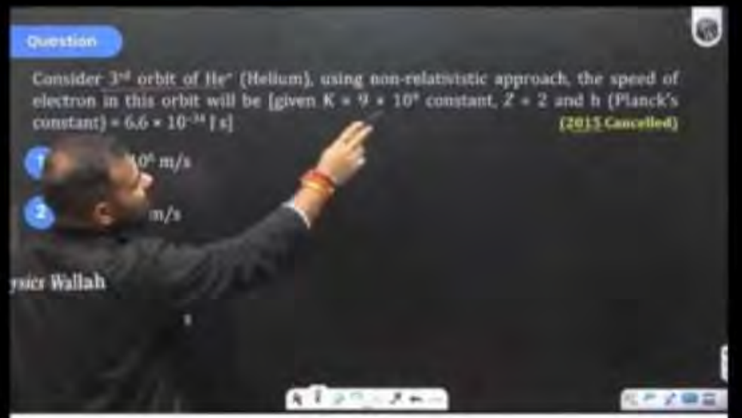
25. A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period 2T. The value of ρ_2/ρ_1 is :

(1) 4

(2) 2

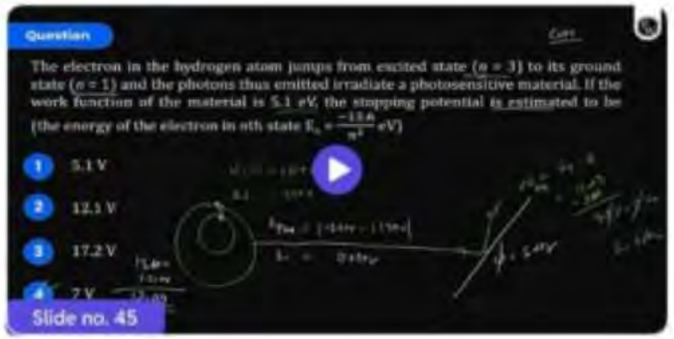
(3) $\frac{1}{2}$

(4) $\frac{1}{4}$

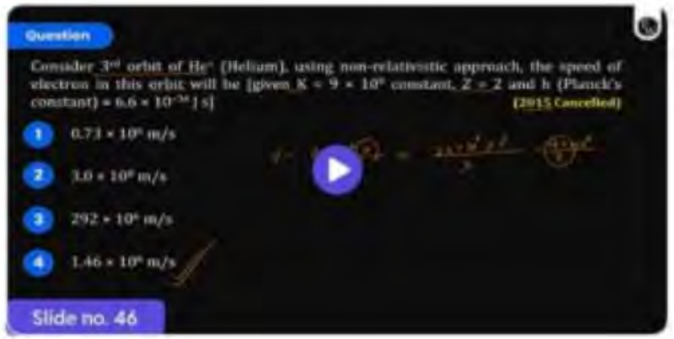


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Slide no. 45



Slide no. 46

40. Consider that an electron is revolving in an excited state of Hydrogen atom with velocity $\sqrt{25.6}\times10^5$ ms^{-1}. The radius of the orbit is $x\times10^{-9}$ m. The value of x is :

(Take the mass of electron to be 9×10^{-31} kg, charge of electron $= -1.6 \times 10^{-19}$ C and $\frac{1}{4\pi\epsilon_0}$ $=9\times10^9$ N m^2 C^{-2}]

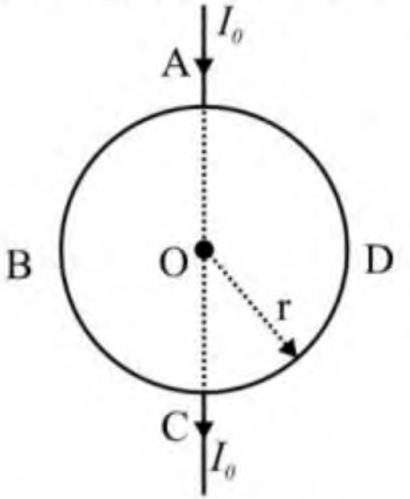
(1) 4 (2) 3

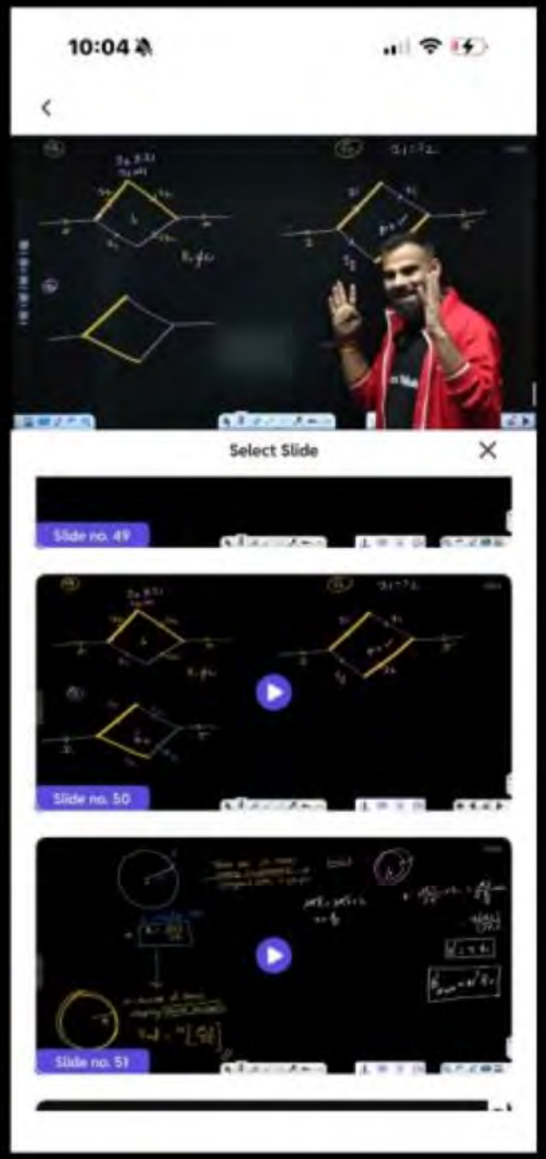
(3) 2 (4) 1

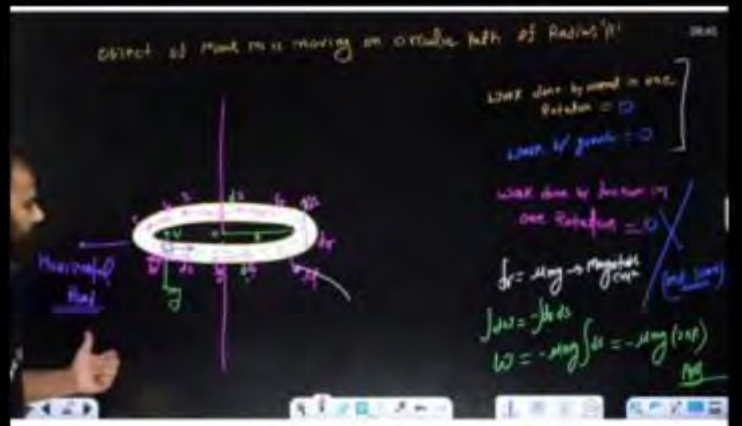
44.



11. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half of ADC. Magnitude of magnetic field at the center O of the loop is:




10:04
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Slide no. 50
Slide no. 51



Select Slide



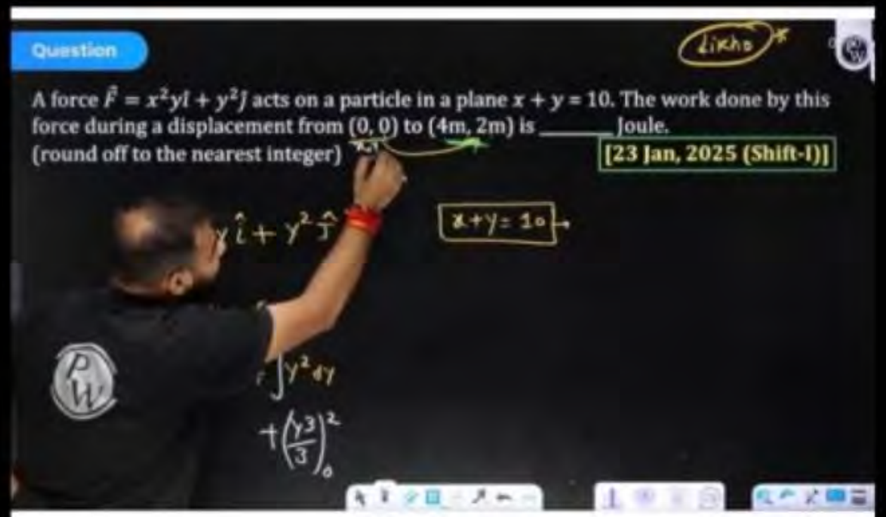




1. A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is:

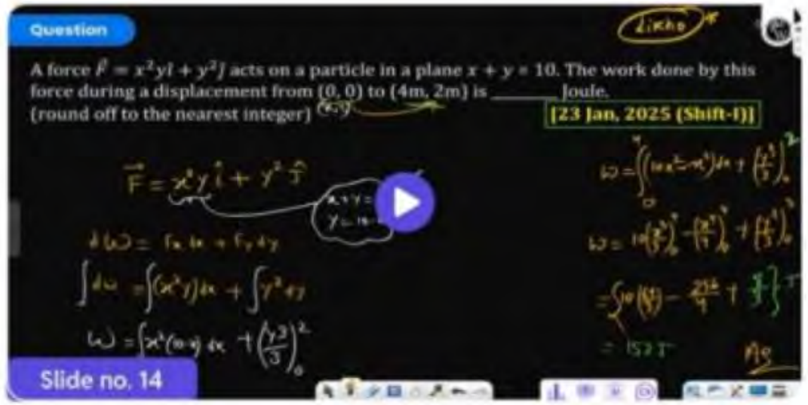
(1) 3 (2) 1

(3) $\frac{1}{3}$ (4) $\frac{1}{2}$

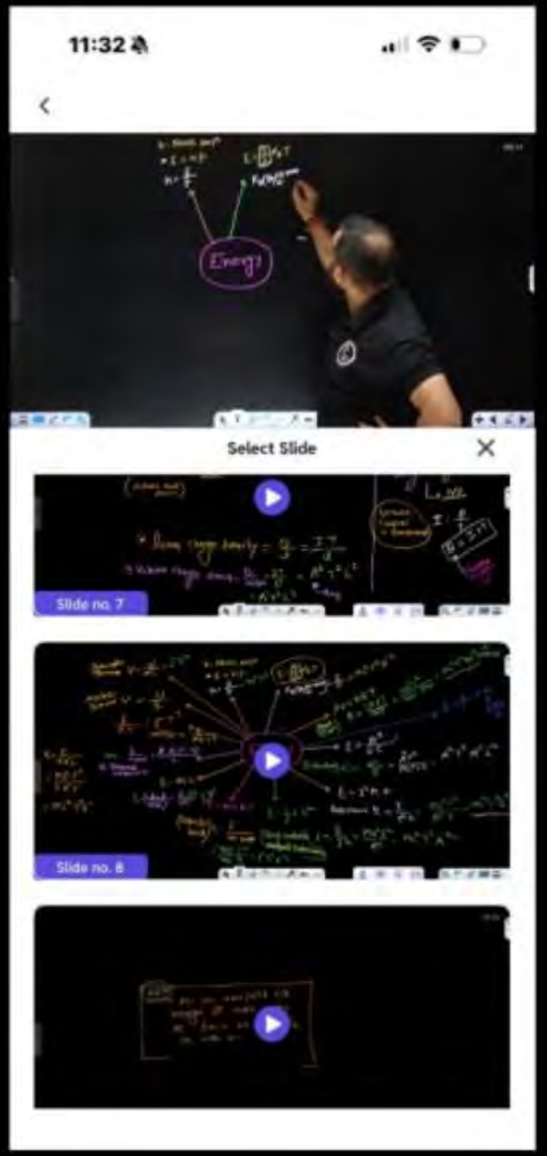


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37. Consider that σ_s, k_B, b represent Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is:

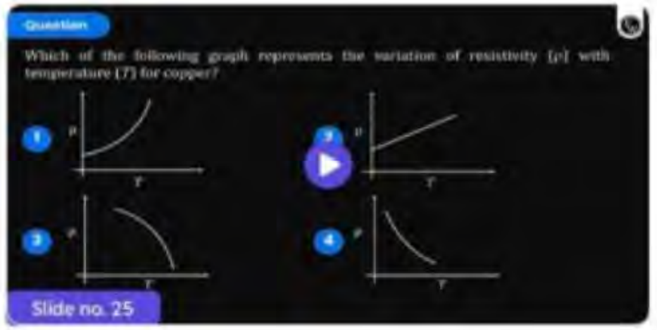
(1) $[L^{-1}T^{-1}K^{-2}]$ (2) $[L^{-1}K^{-2}]$

(3) $[L^{-1}T^{-1}K^{-3}]$ (4) $[L^{-1}T^{-1}K^{-4}]$



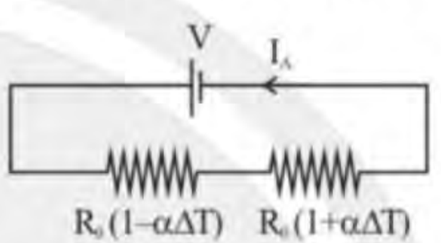
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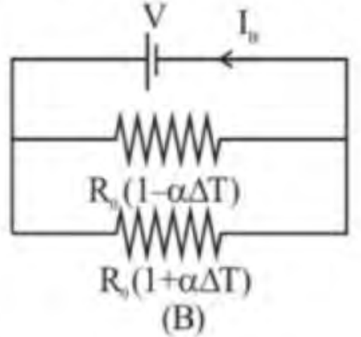




36. Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature of coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is :



(A)



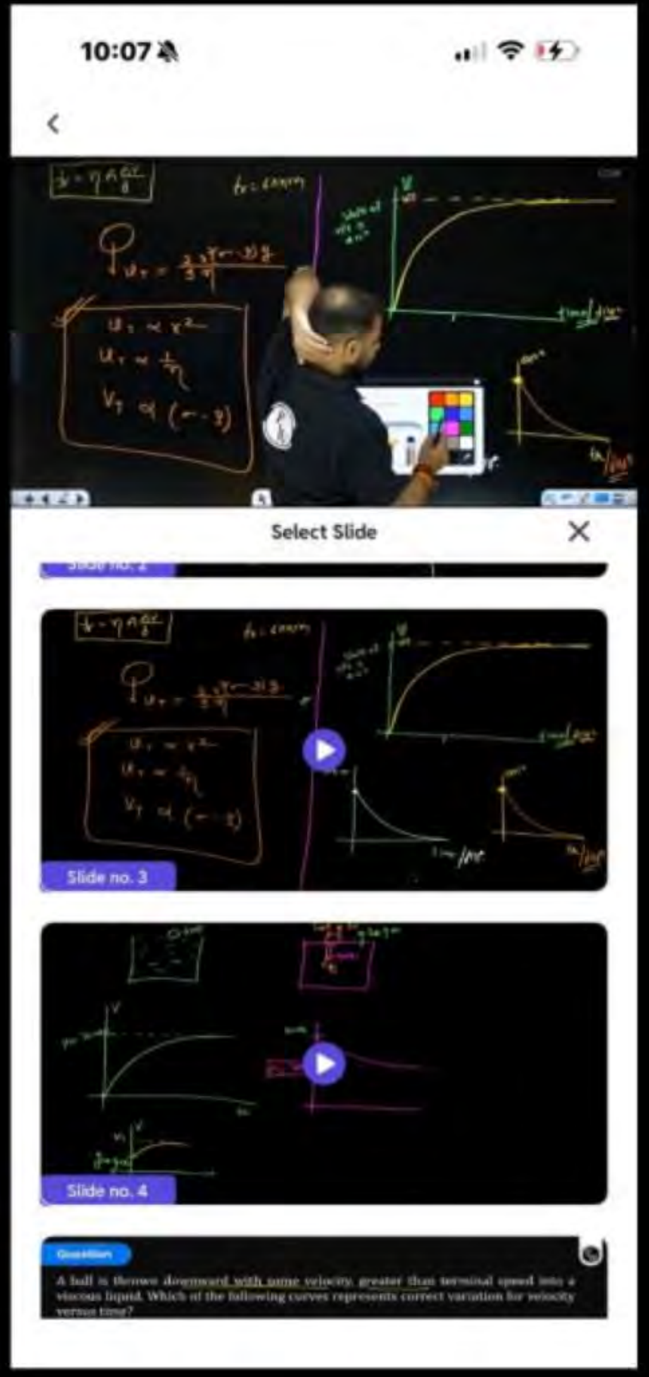
(B)

(1) I_A remains constant while I_B increases

(2) I_A decreases while I_B increases

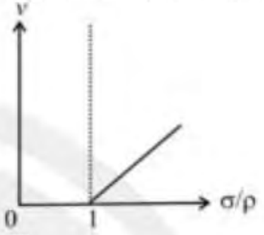
(3) I_A increases while I_B decreases

(4) both I_A and I_B remain constant

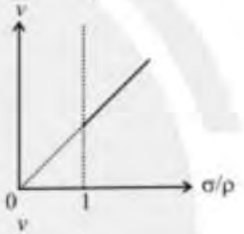


12. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by:

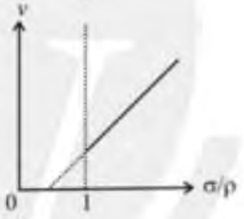
(1)



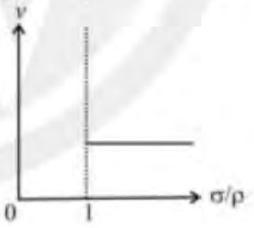
(2)

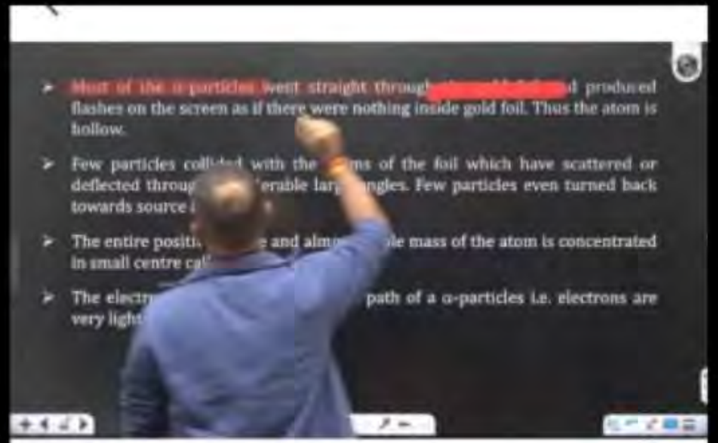


(3)

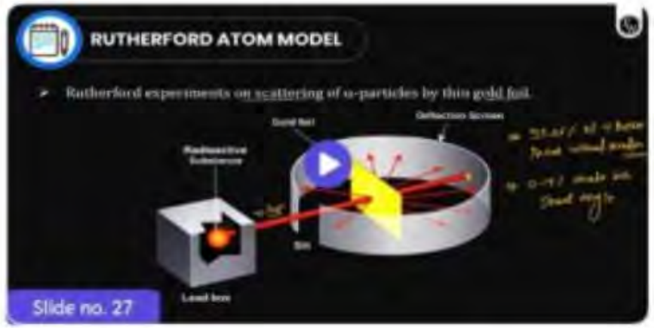


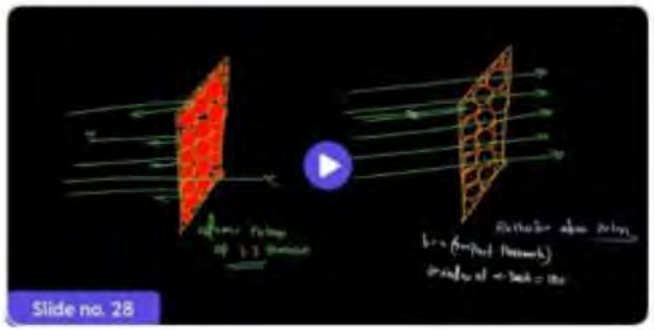
(4)





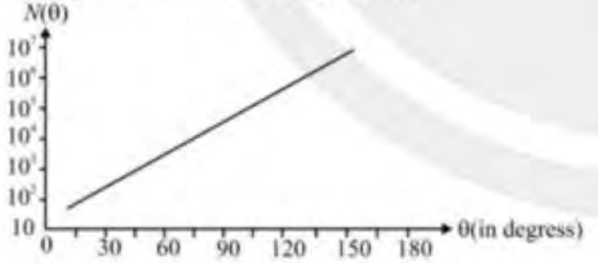
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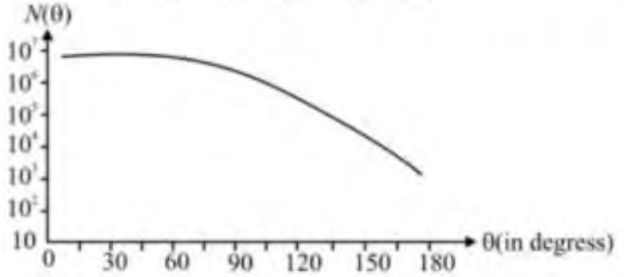


35. In Geiger-Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the **correct** plot?

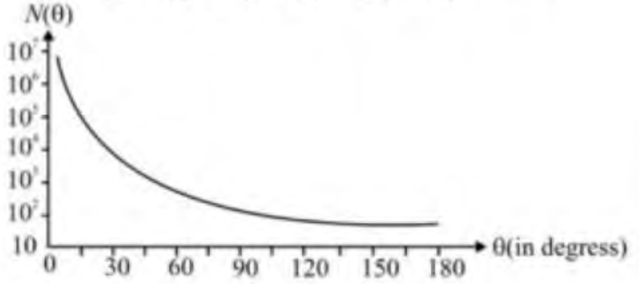
(1)



(2)



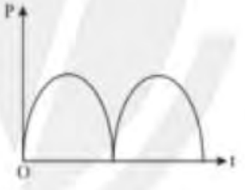
(3)





32. A conducting loop of finite resistance lies on the x - y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is :

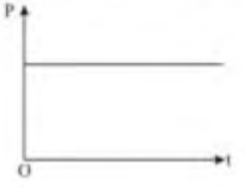
(1)



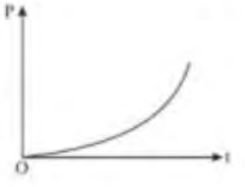
(2)



(3)

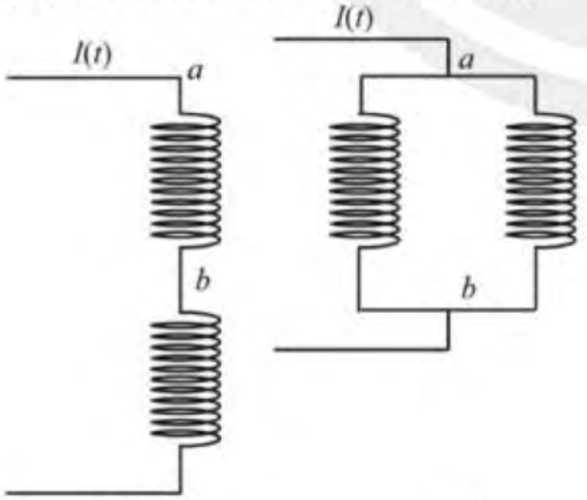


(4)



3. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P / E_Q is :

[Neglect the effect of mutual inductance.]



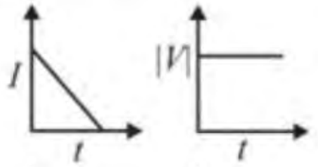
(P)

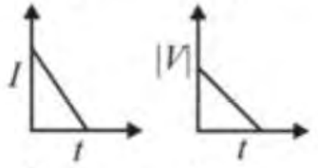
(Q)

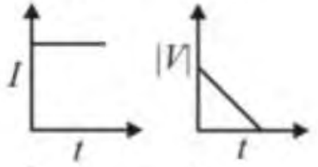
(1) 1/4
(2) 1/2
(3) 1
(4) 2

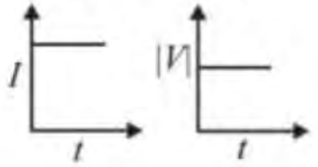


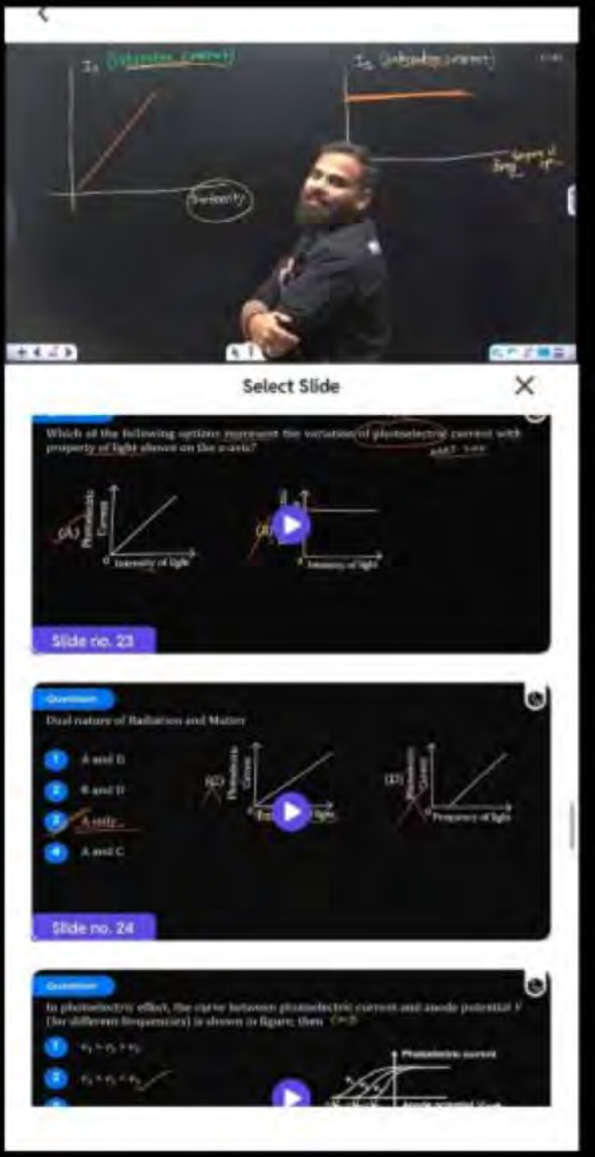
8. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by :

(1) 

(2) 

(3) 

(4) 



(4) $V_1 < V_2, V_3 = 0$

19. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is :

(Take speed of light = 3×10^8 ms^{-1}, charge of electron $= -1.6 \times 10^{-19}$ C and mass of electron $\frac{1}{4\pi \in_0} = 9 \times 10^{-31}$ kg)

(1) $\frac{2}{450}$ (2) $\frac{1}{250}$

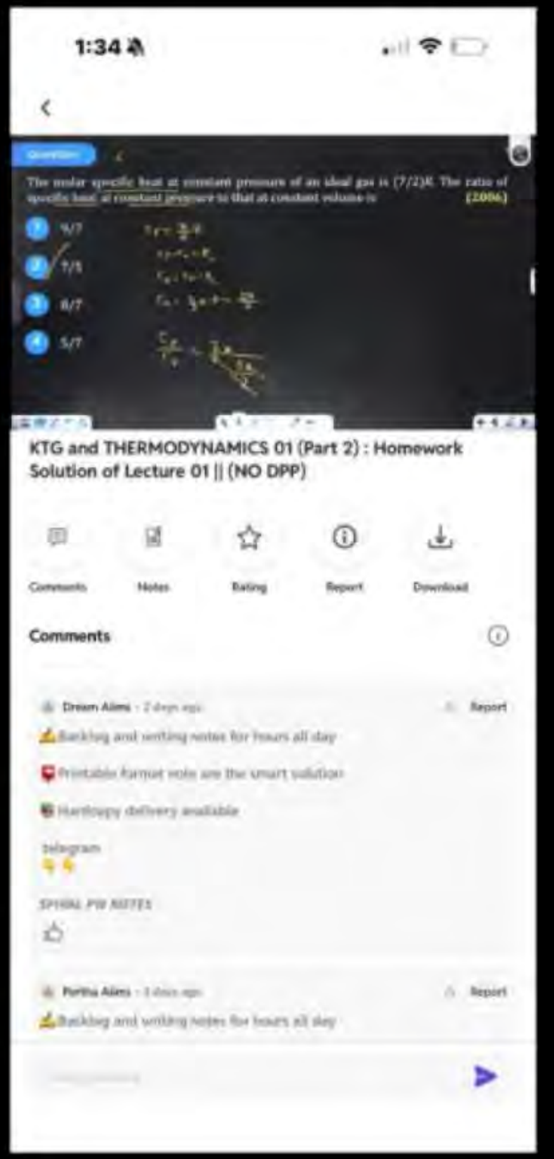
(3) 225 (4) 275



45. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities $\frac{C_p}{C_v}$ of the gas is $\frac{8}{7}$, then the value of f is:

(1) 4 (2) 3

(3) 2 (4) 1



Thank You